

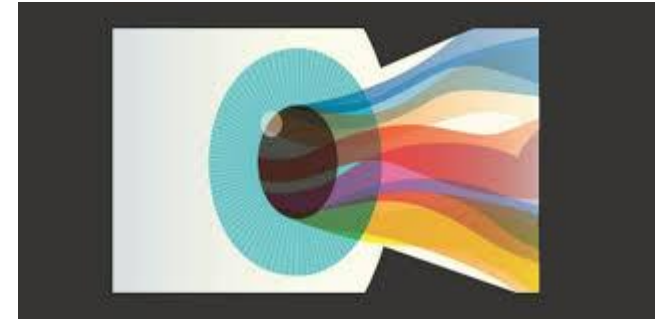
Fundamental Concepts

Human Capabilities

- Our Senses
 - How to sense changes/information
- Our Cognition
 - How we process and interpret input
- Our Motor System
 - How can we react to input and cognition

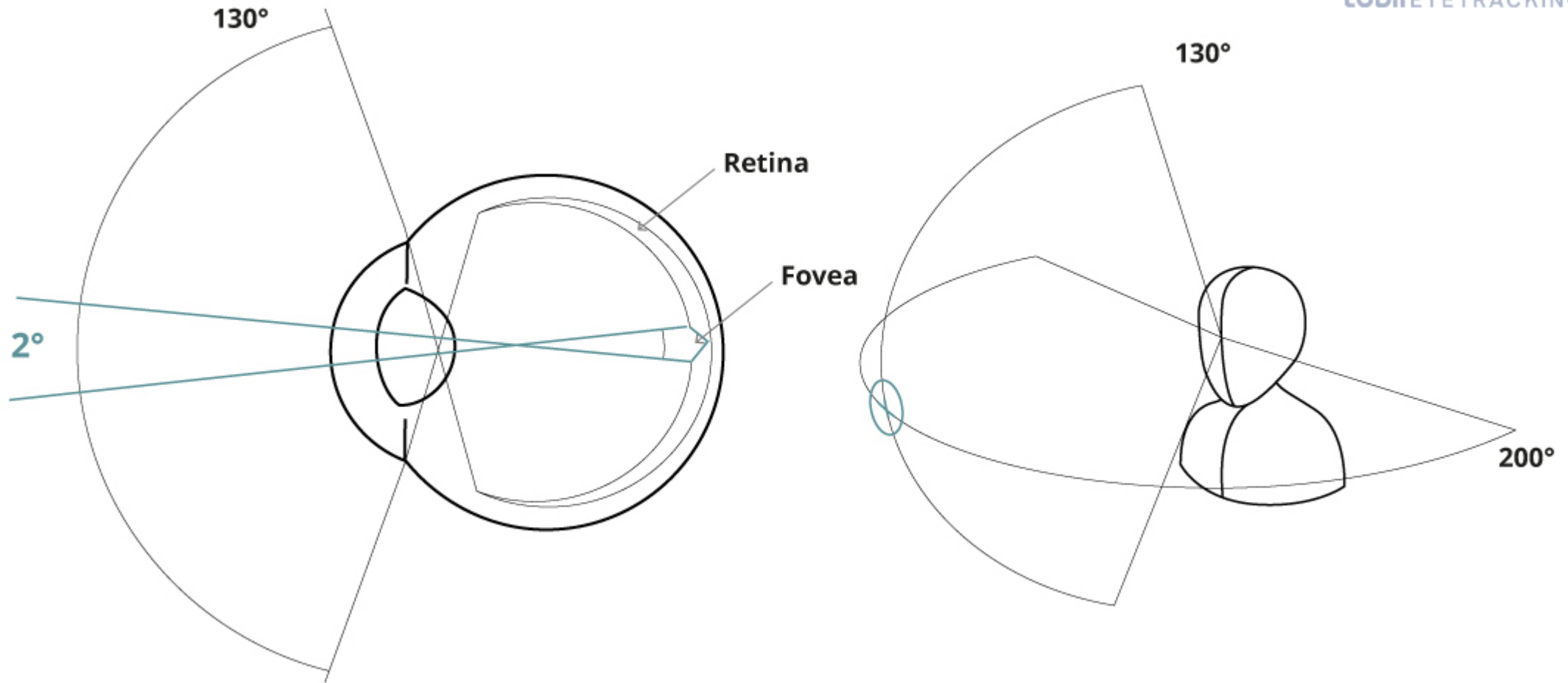
Human Senses

- Sight, hearing, touch important for the design of current interfaces
 - Others: smell, tastes, kinesthetic
- Abilities and limitations constrain design space

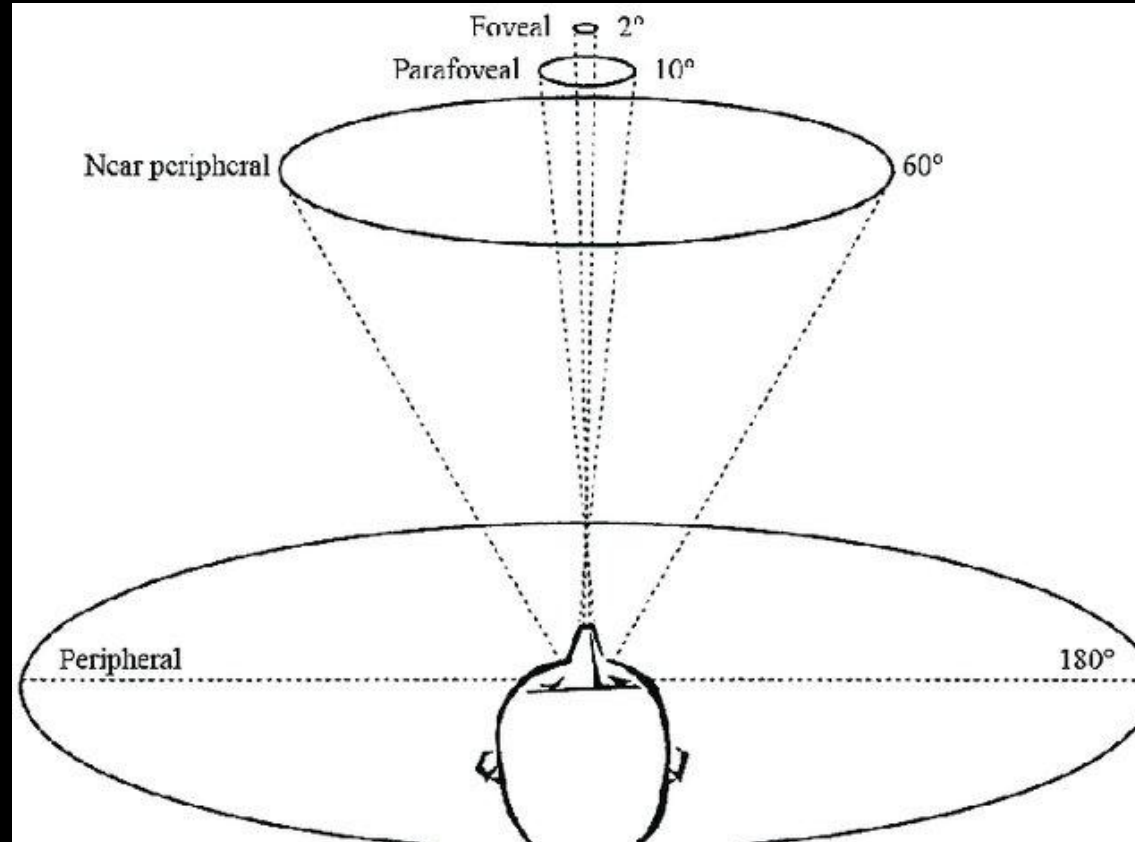


Human Vision – View Frustum

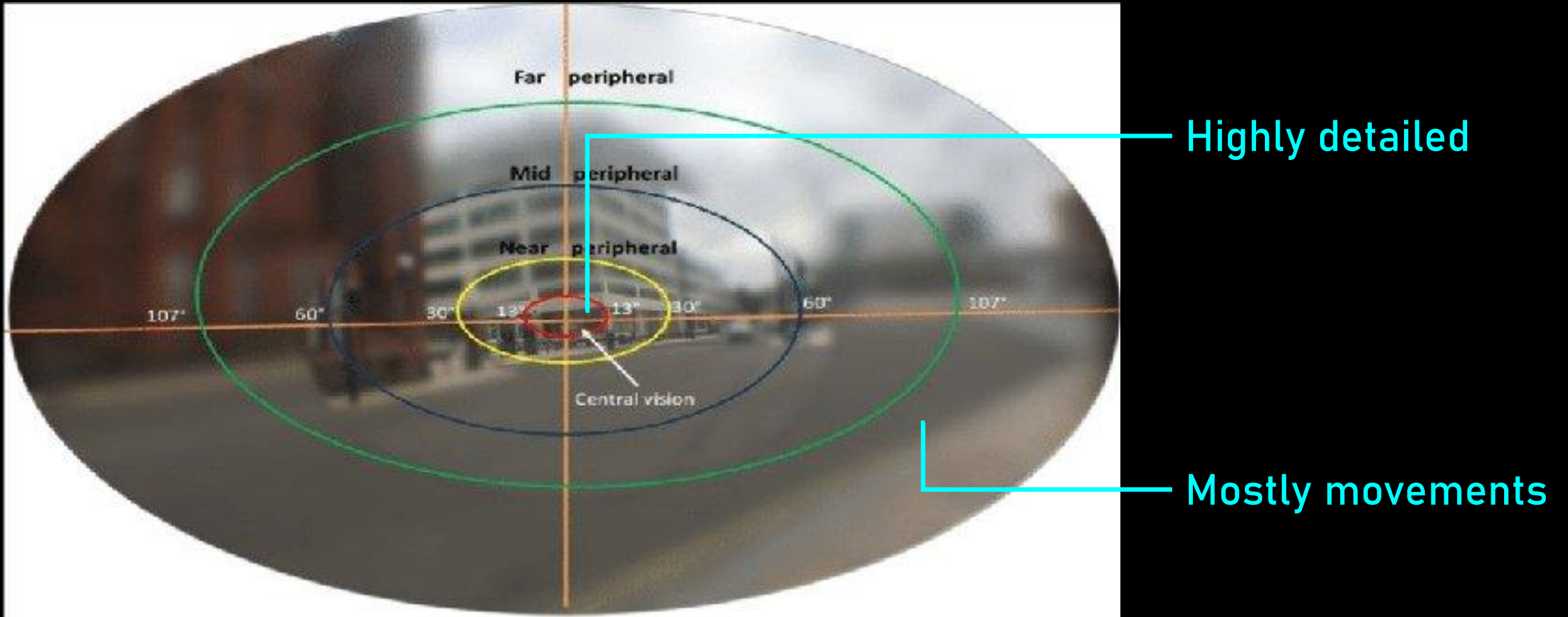
tobii EYETRACKING



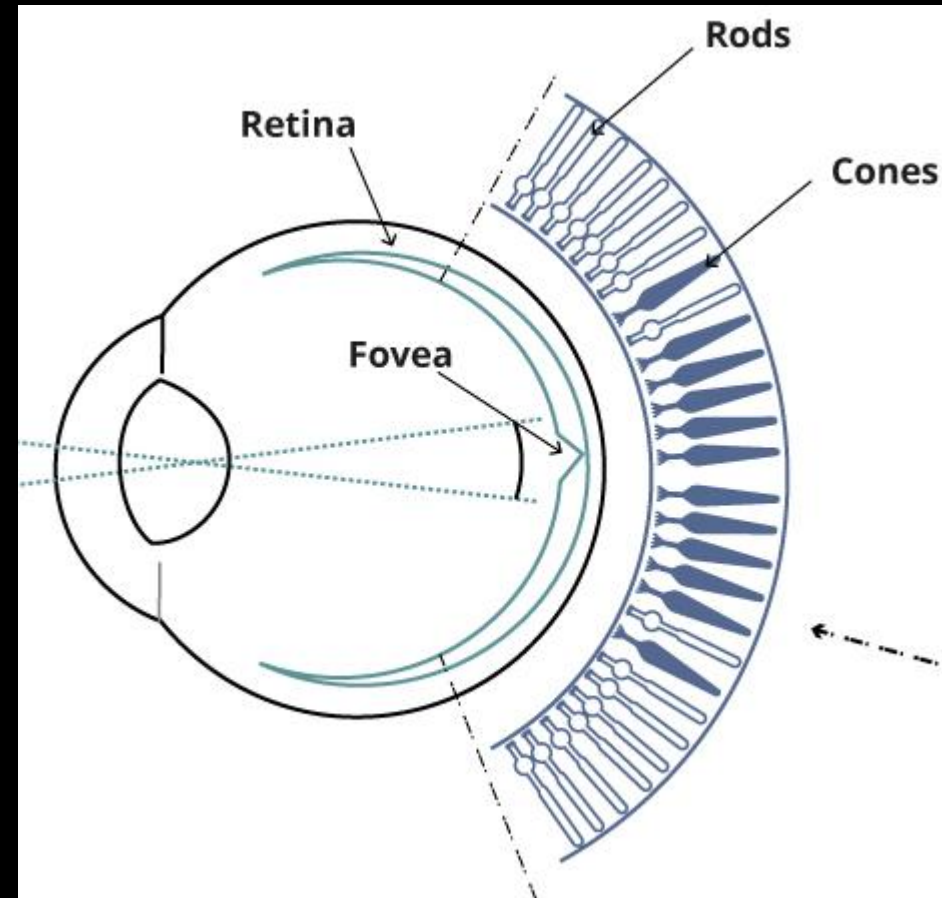
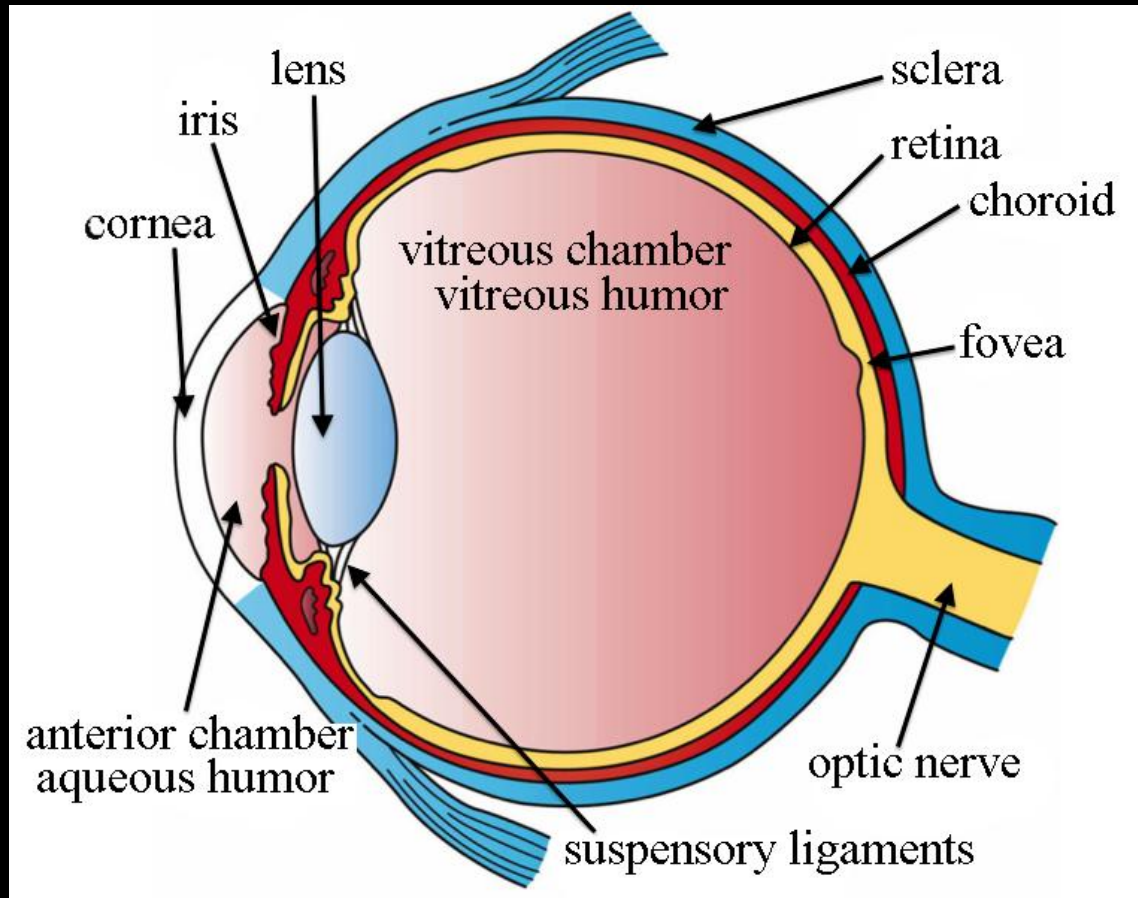
Human Vision – View Frustum



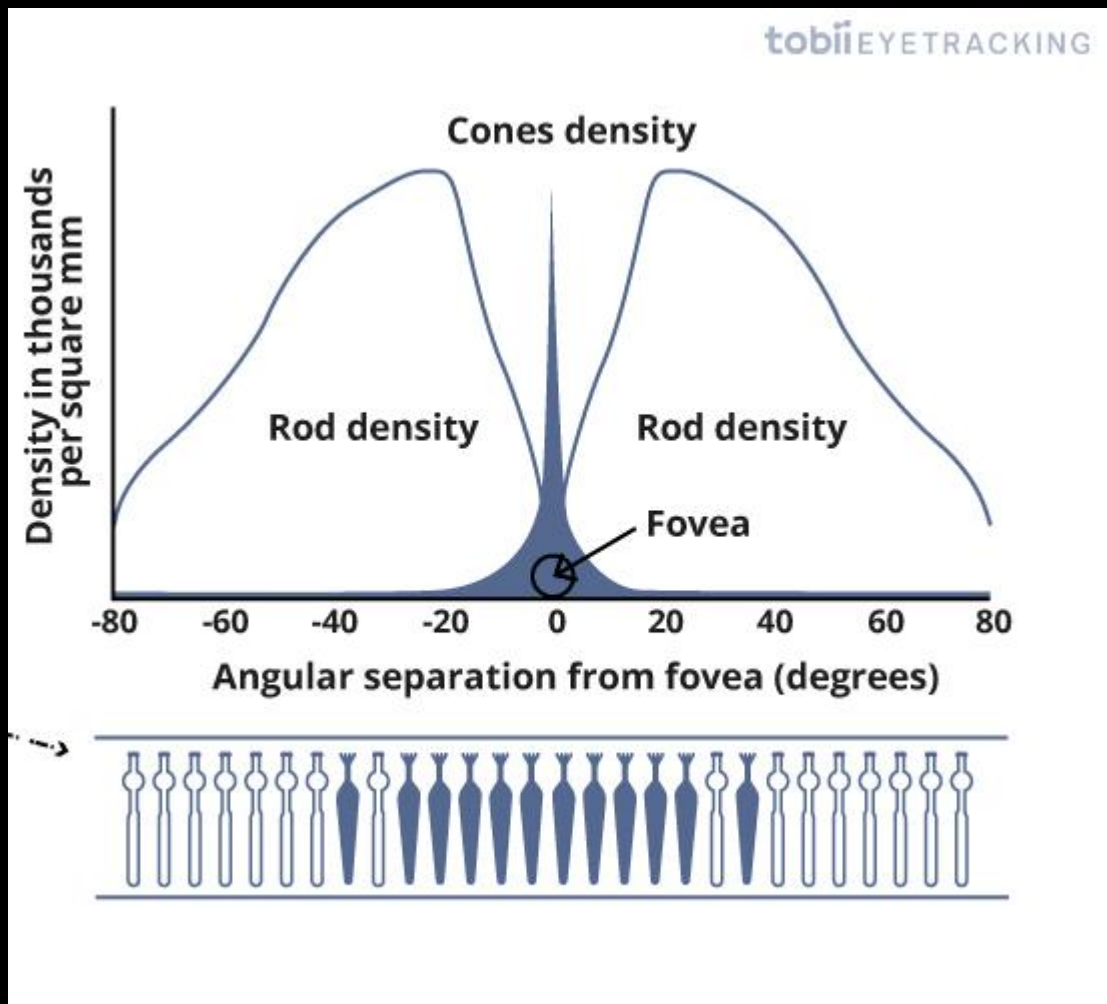
Human Vision – View Frustum



Human Vision – Color Perception

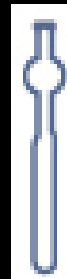


Human Vision – Color Perception



Rods

- 120 millions
- B/W
- 1000x more sensitive than cones



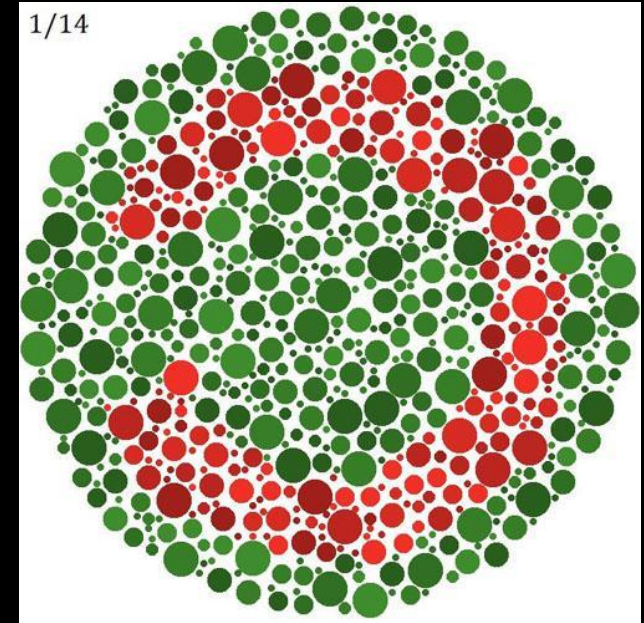
Cones

- 6-7 millions
- 60% R – 30% G – 10% B

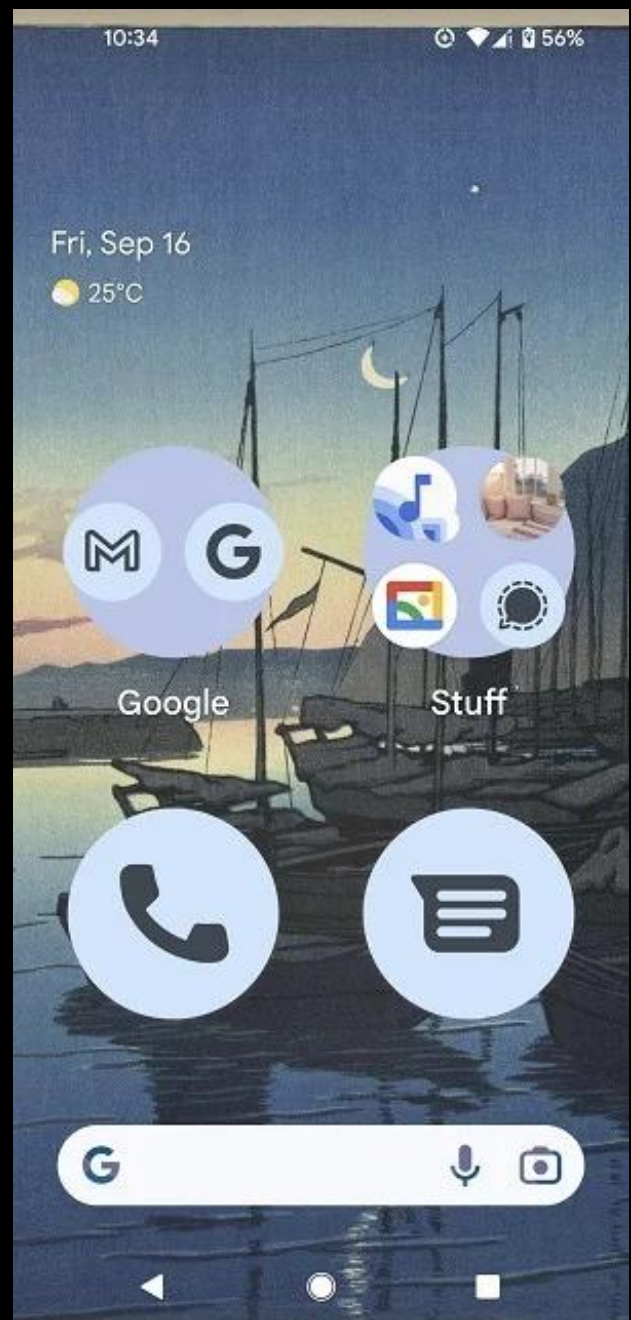
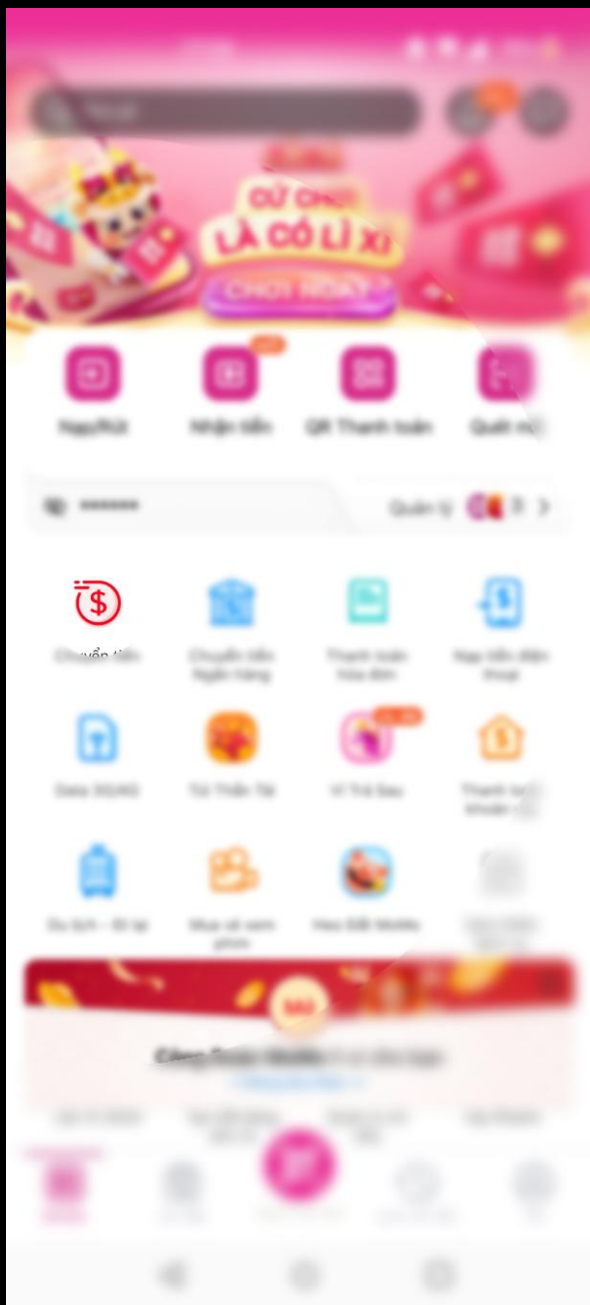


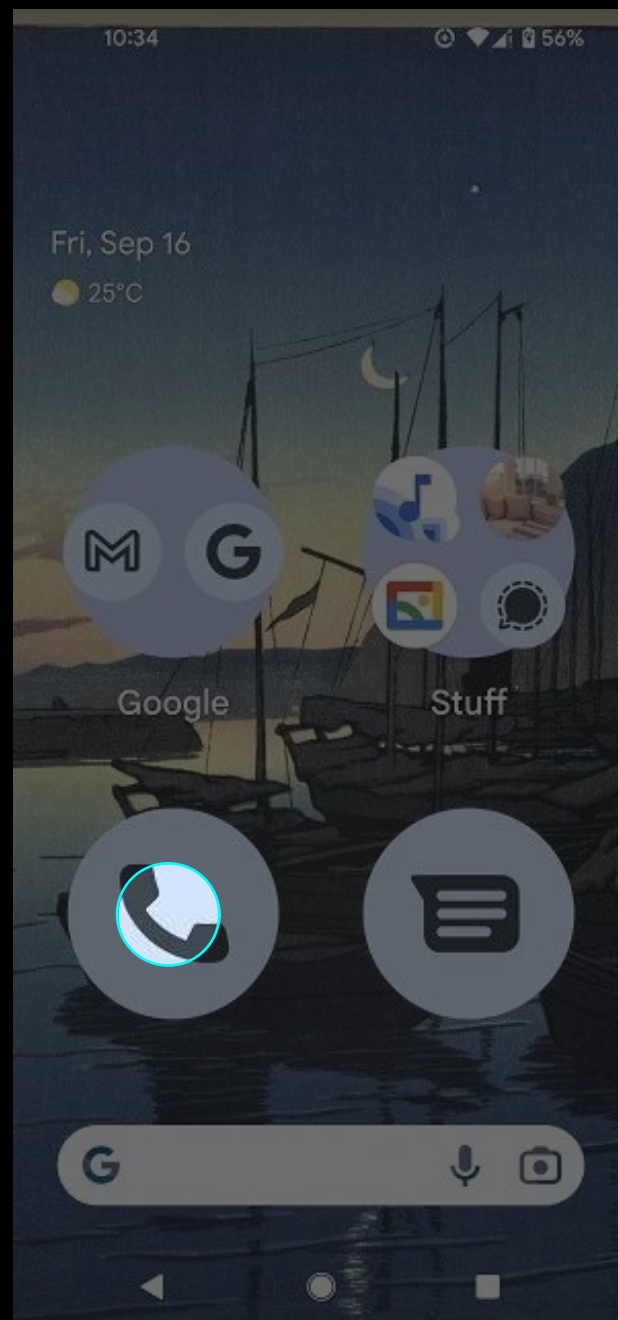
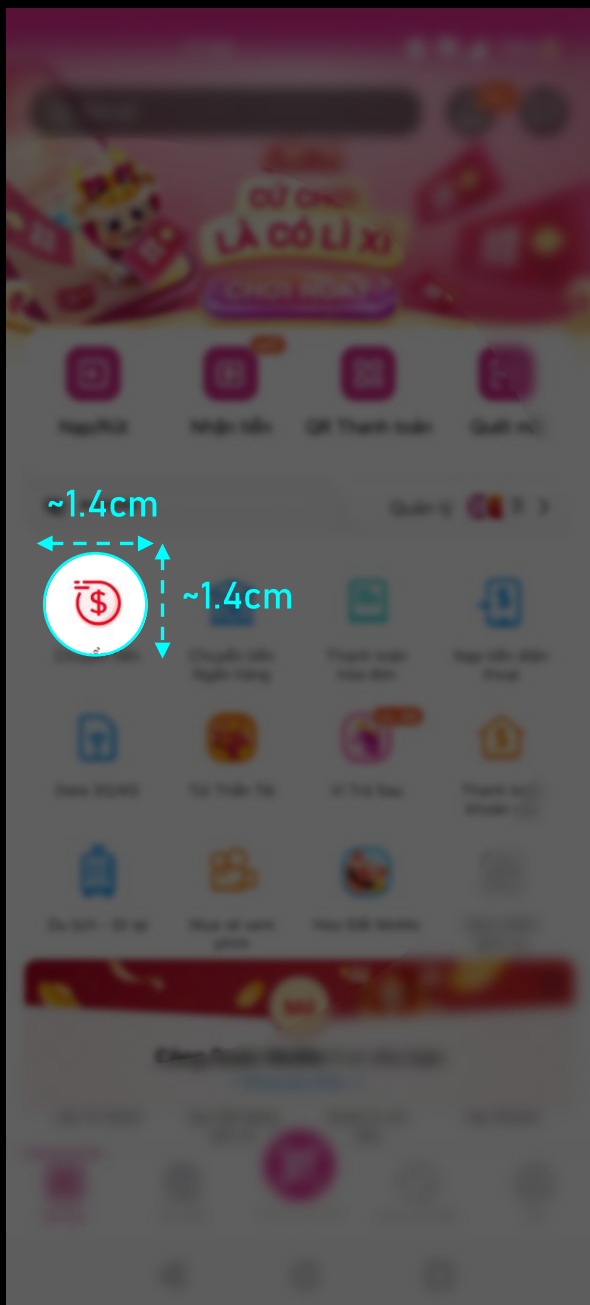
Human Vision

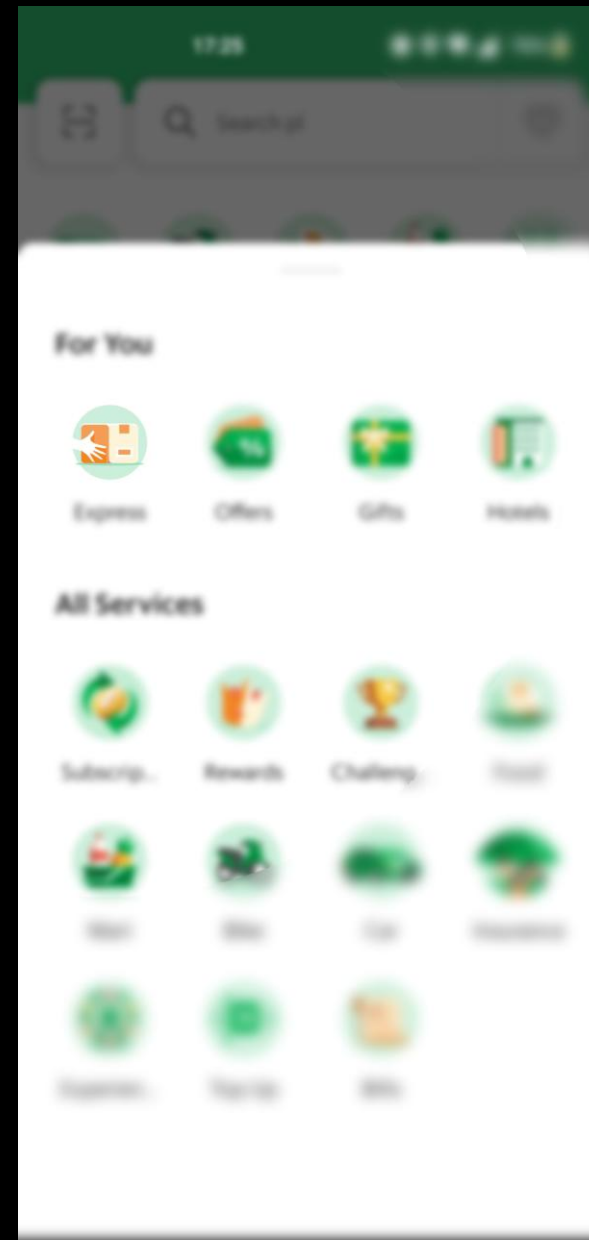
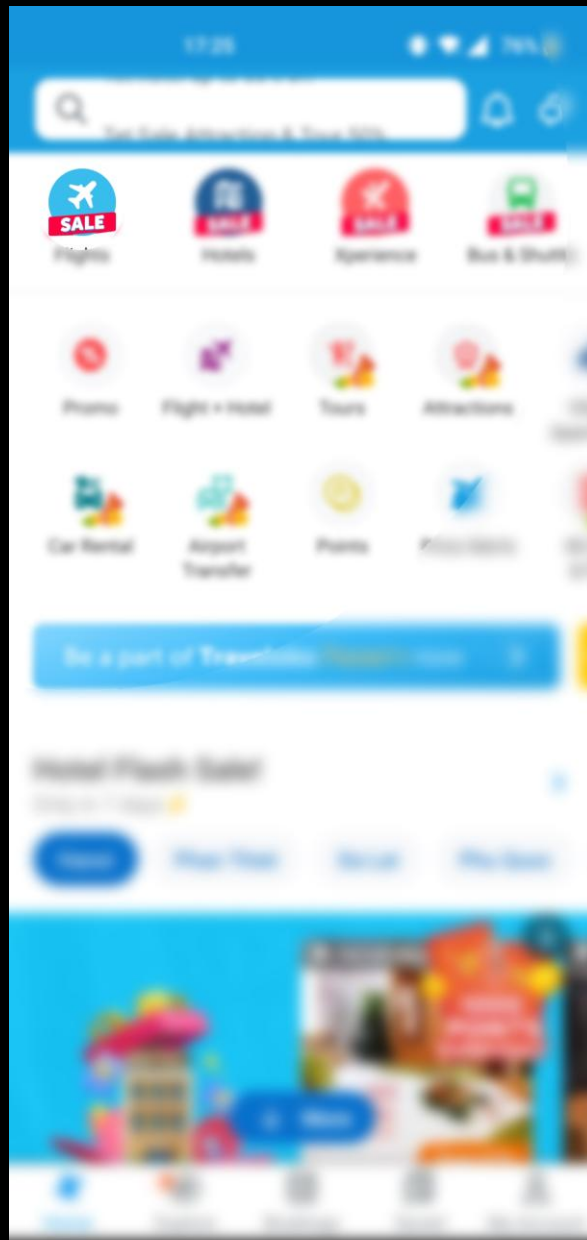
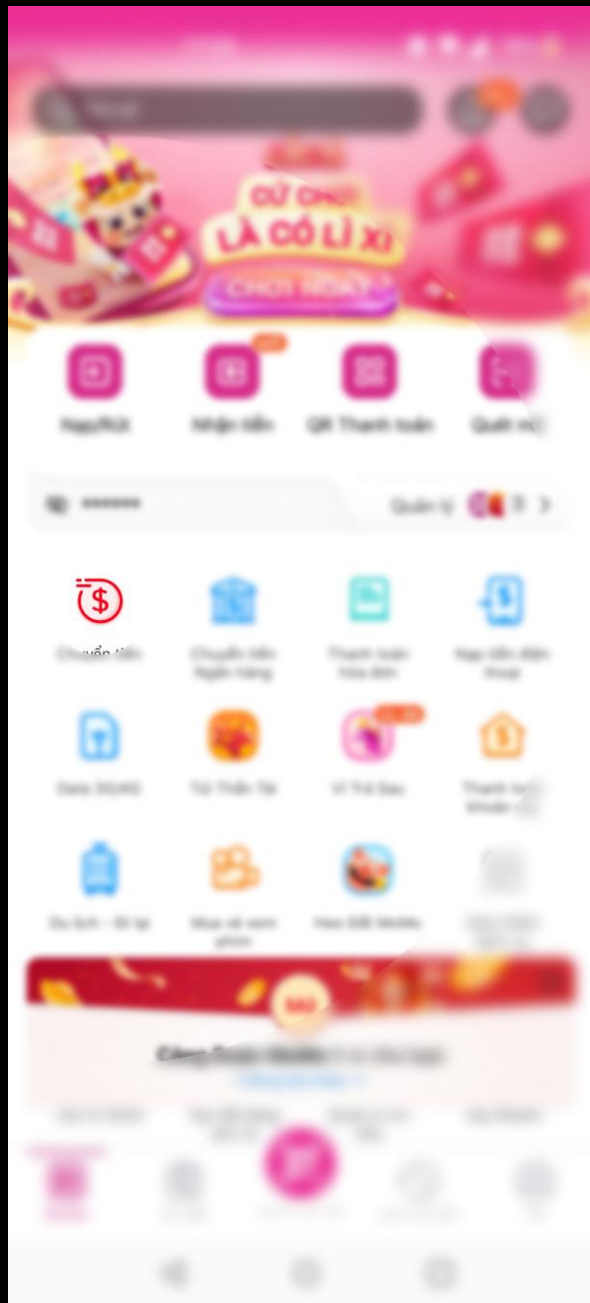
- Color Perception
 - 7-8% of males cannot distinguish red from green
 - 0.4% of women
- Stereopsis
 - Monocular (color, brightness, shape)
 - Binocular (size, interposition, perspective, parallax)













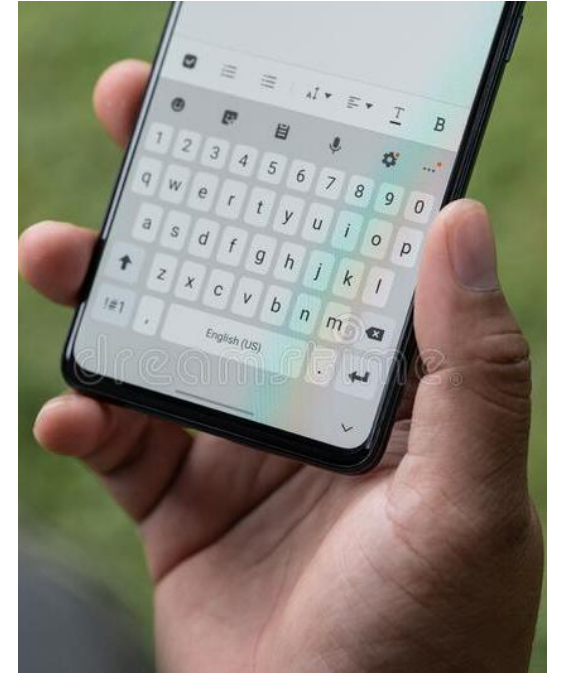
Audition (Hearing)

- Capabilities:
 - Pitch – frequency (20 – 20,000 Hz)
 - Loudness – amplitude (30 – 100 dB)
 - Location (5° source & stream separation)
 - Timbre – type of sound (music instruments)



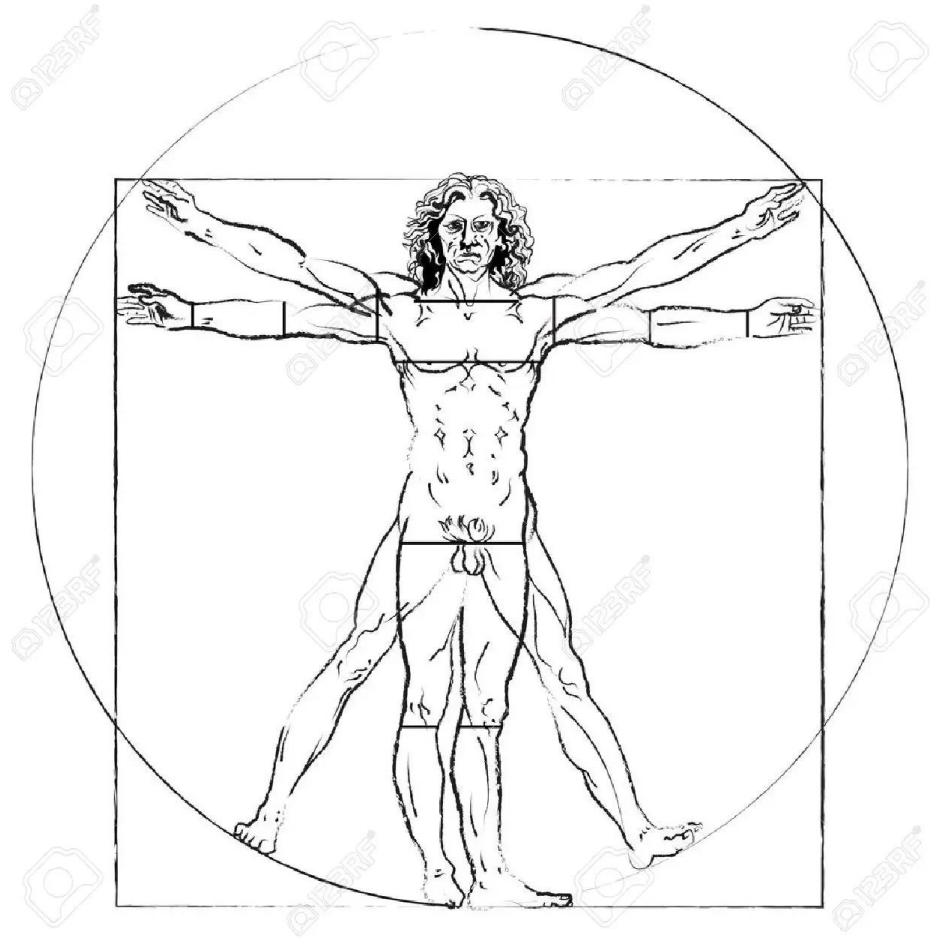
Touch

- Three main sensations handled by different types of receptors
 - Pressure (normal)
 - Intense pressure (heat/pain)
 - Temperature (hot/cold)
- Sensitivity, Dexterity, Flexibility, Speed
- Where important?
 - Mouse, Keyboards, Buttons, VR, Surgery



Motor System (Our Output System)

- Capabilities
 - Range of movement, reach, speed, strength, dexterity, accuracy
 - ⇒ Workstation design, device design
- Often cause of errors
 - Wrong button
 - Double-click vs. Single Click
 - Not-detected mid-air gestures
- Principles
 - Feedback is important
 - Minimize eye movement



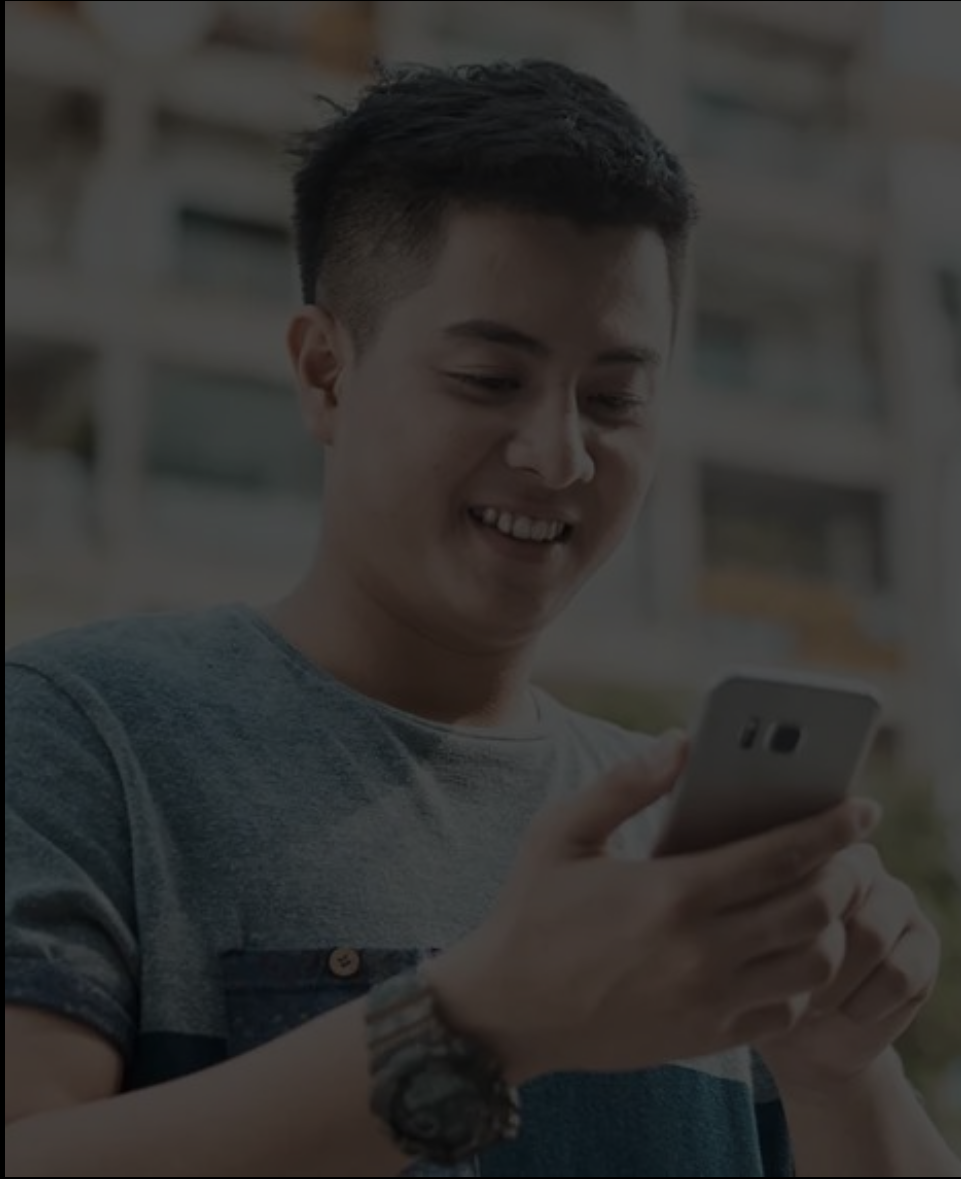


<https://www.youtube.com/watch?v=2YduyTzZews>

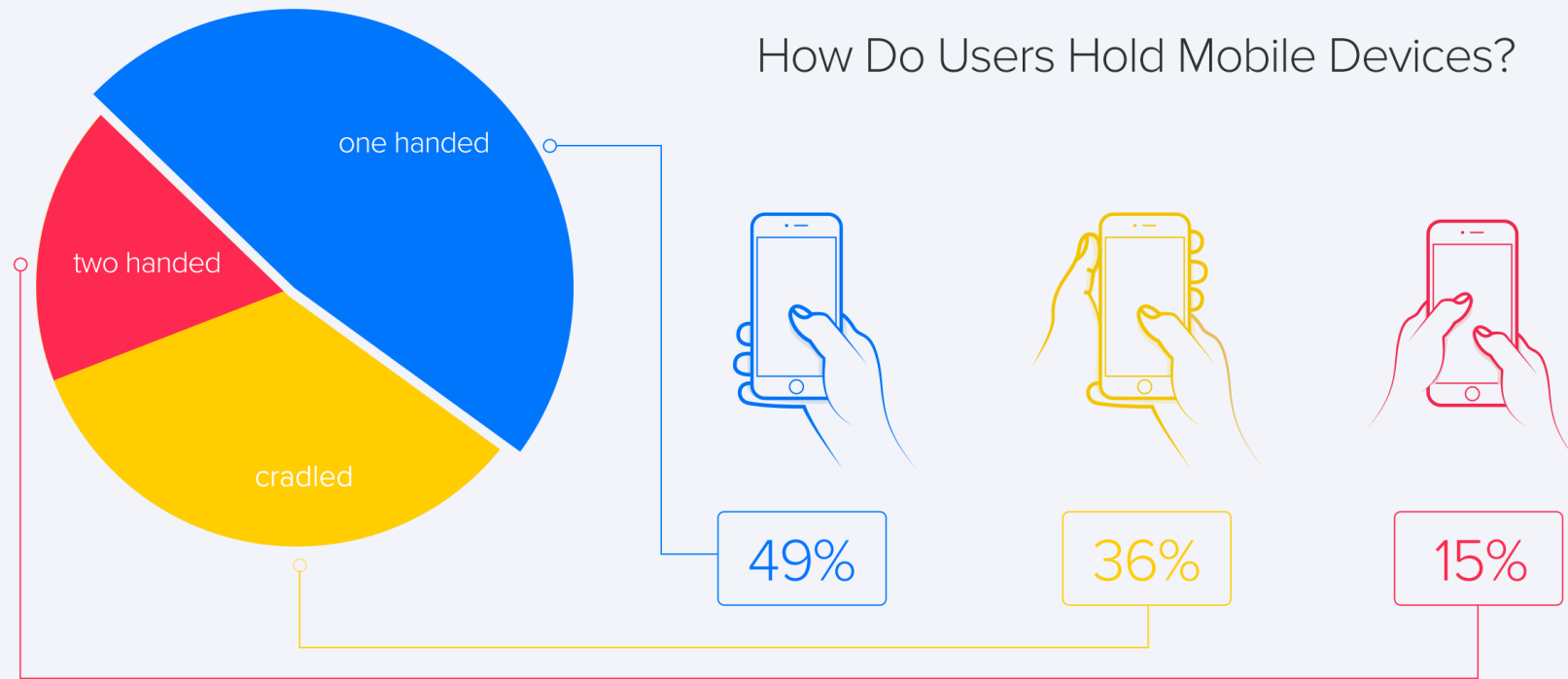


How to **Use AssistiveTouch on Apple Watch**





LYFT'S USER INTERACTION INSIGHTS

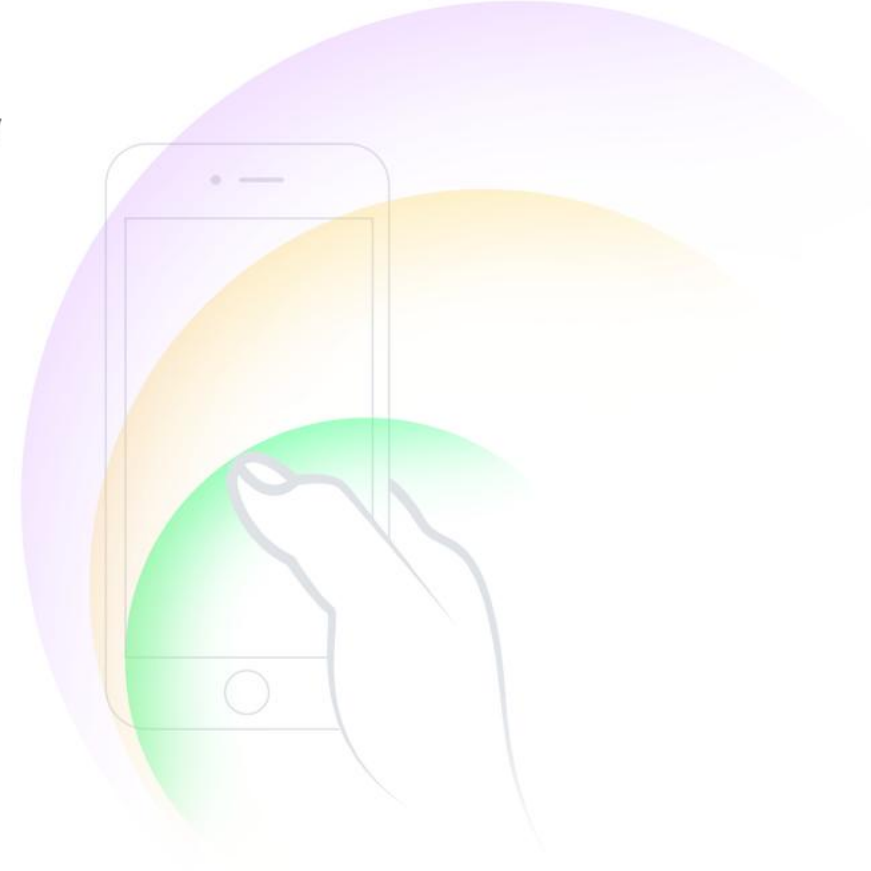
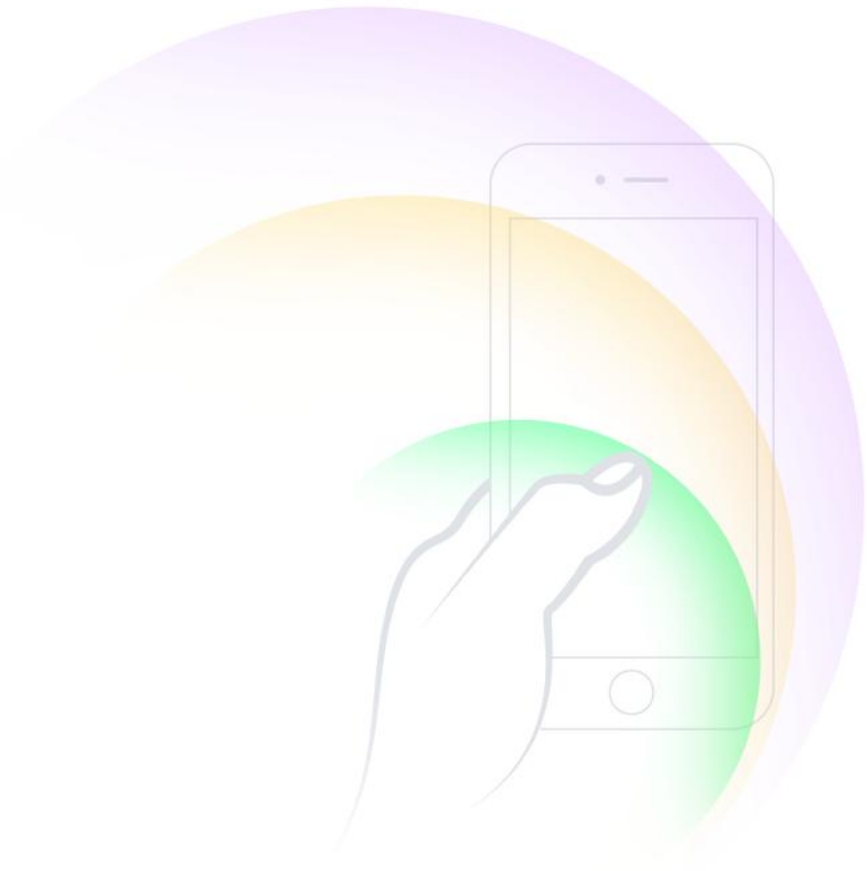


LYFT'S USER INTERACTION INSIGHTS

Reachability

Left-Handed & Right- Handed Example

- Good
- Medium
- Out of Reach (OW)

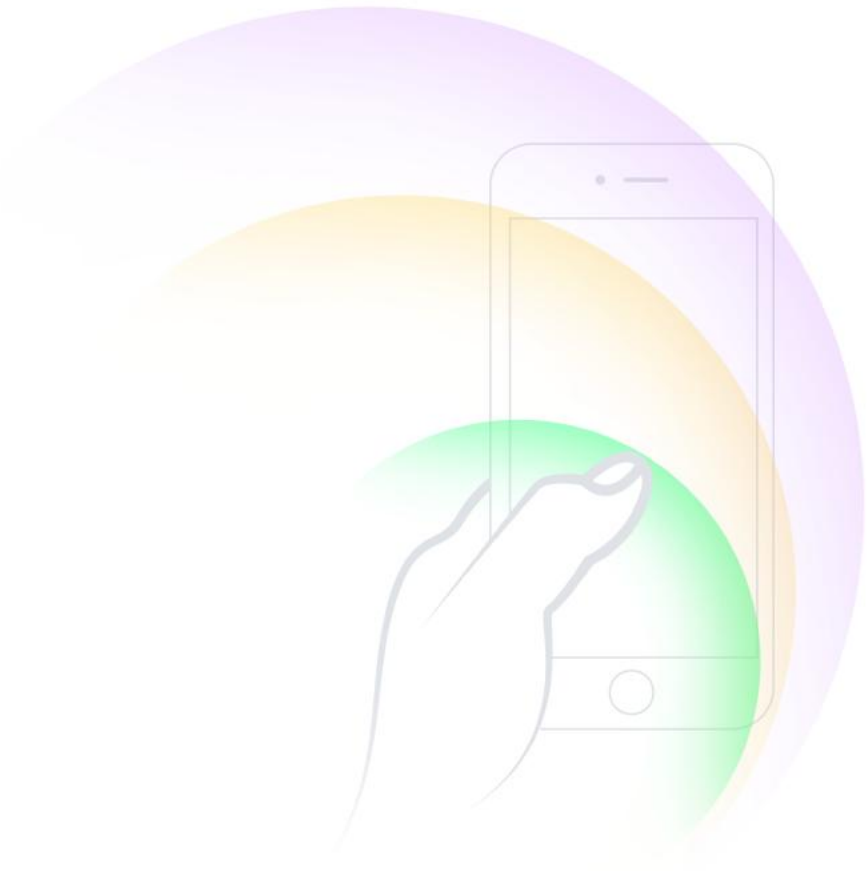


LYFT'S USER INTERACTION INSIGHTS

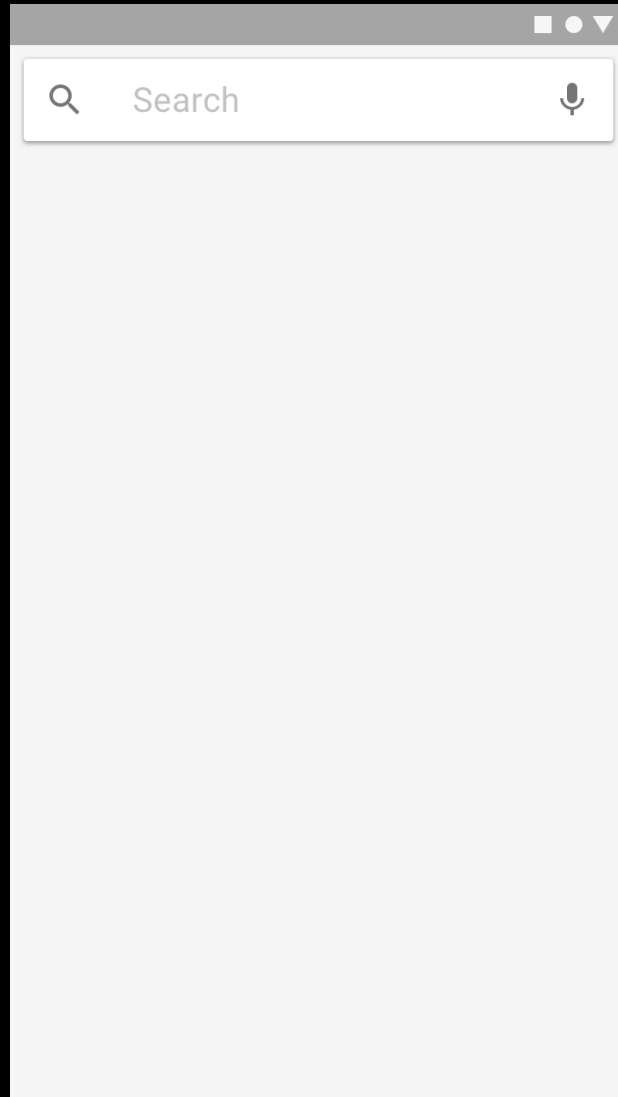
Reachability

Left-Handed & Right- Handed Example

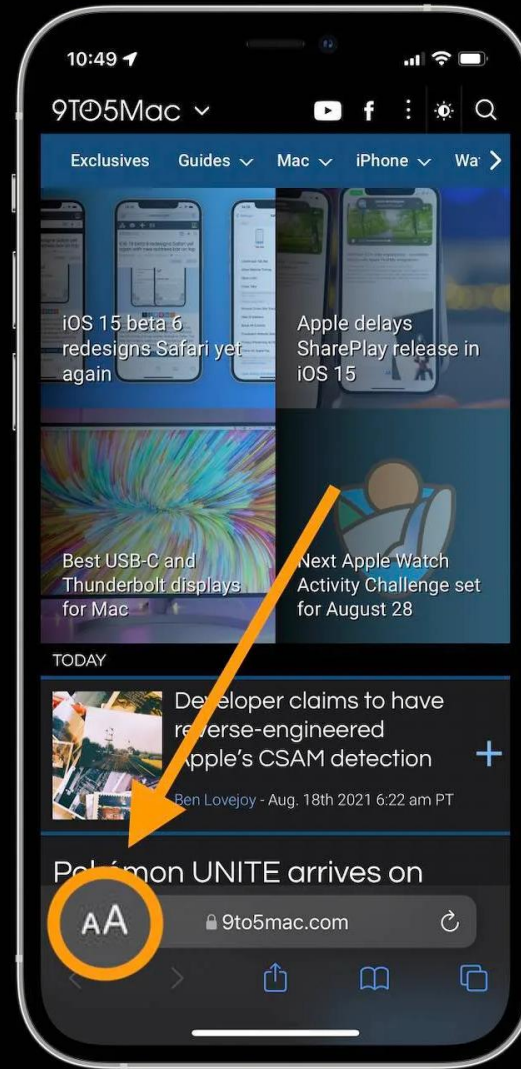
- Good
- Medium
- Out of Reach (OW)



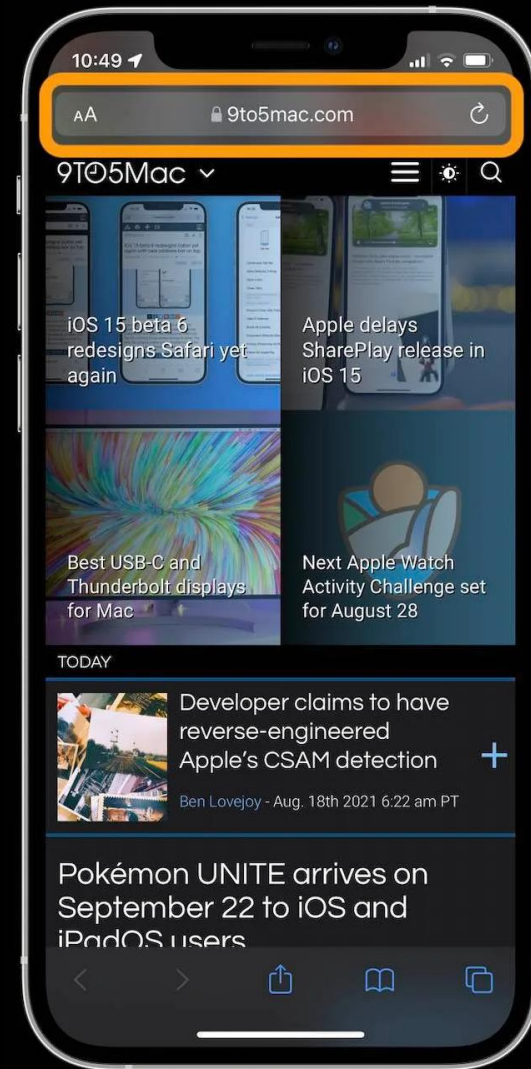
SEARCH BAR ACCORDING TO POPULAR DESIGN GUIDELINES



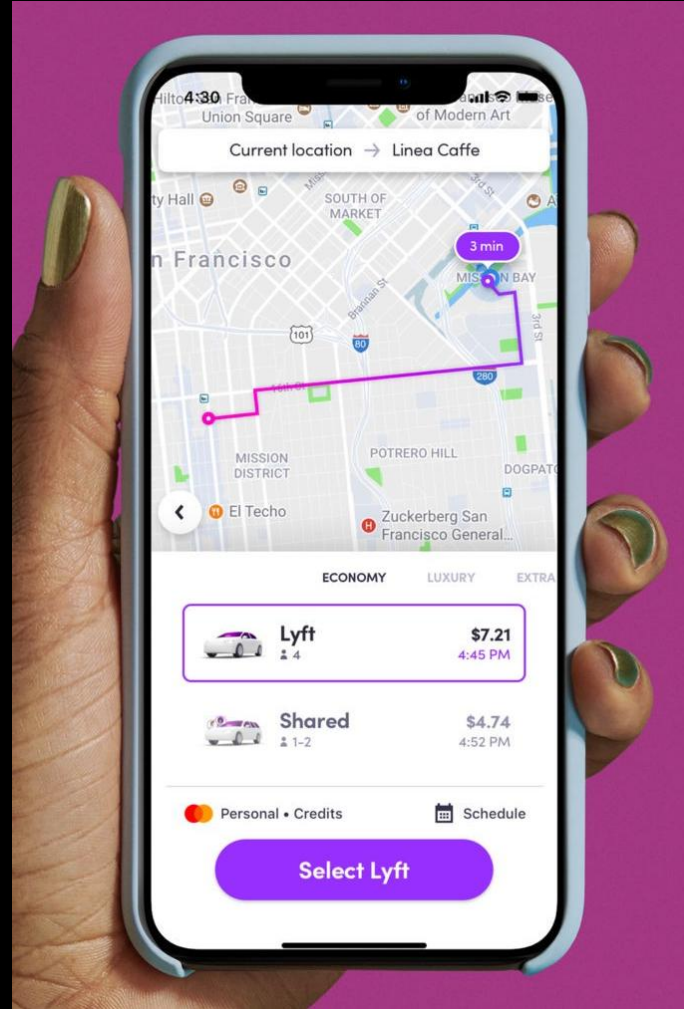
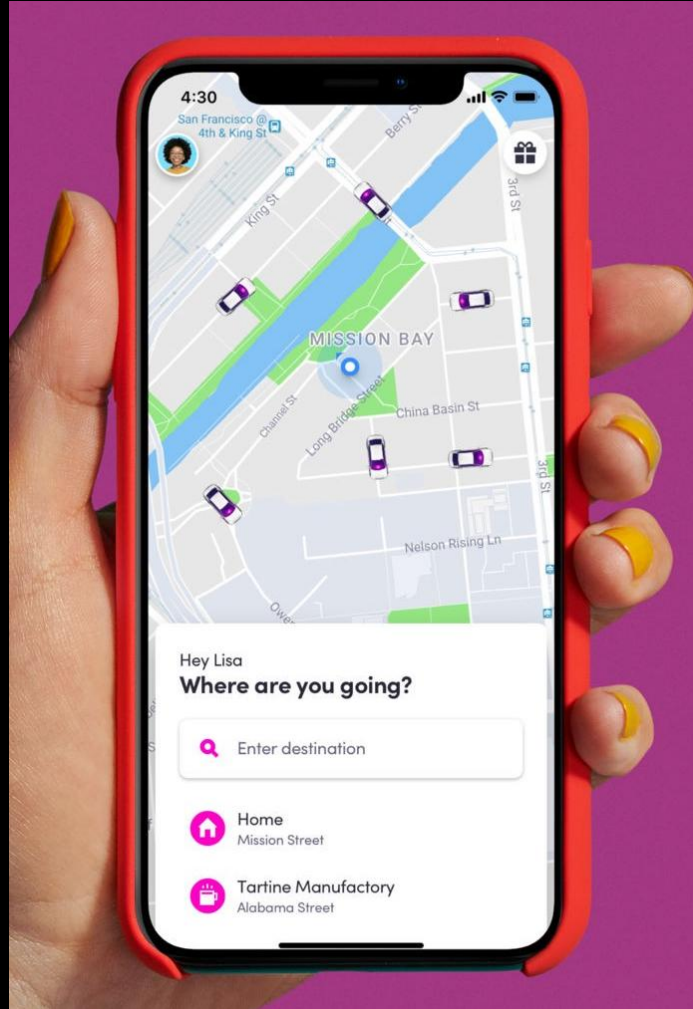
Google Material Design



iOS Material Design



LYFT'S UNCONVENTIONAL RESEARCH BAR



Human Cognition

Attention



Different types of attention

- Selected attention
- Focused attention
- Divided attention

Selected attention



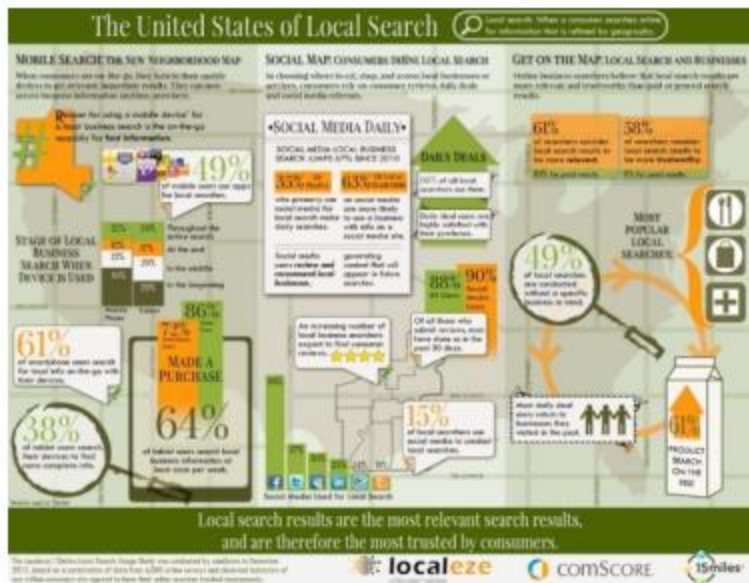
Focused attention



Divided attention



Design implications for attention





Create post



Lê Khánh Duy

Friends ▼

What's on your mind, Duy?



Add to your post



Post

hình dịch trên những ngữ liệu thu thập được, đồng thời xây dựng website hỗ trợ chuyển tự tự động từ chữ Nôm sang chữ Quốc ngữ (đã nộp hồ sơ đăng ký Quyền tác giả).

81% 🔋

12:51 Monday, Feb 13



Vasteras2021



Mobile data
4G



Bluetooth



Flashlight



Auto rotate



Power saving mode
1 day, 12 hr, 56 min left



Airplane mode



Location



Do Not Disturb



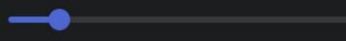
NFC



Nearby Share



Personal hotspot



MobiFone



VPN



AirVisual



FitCoach



Authentic...



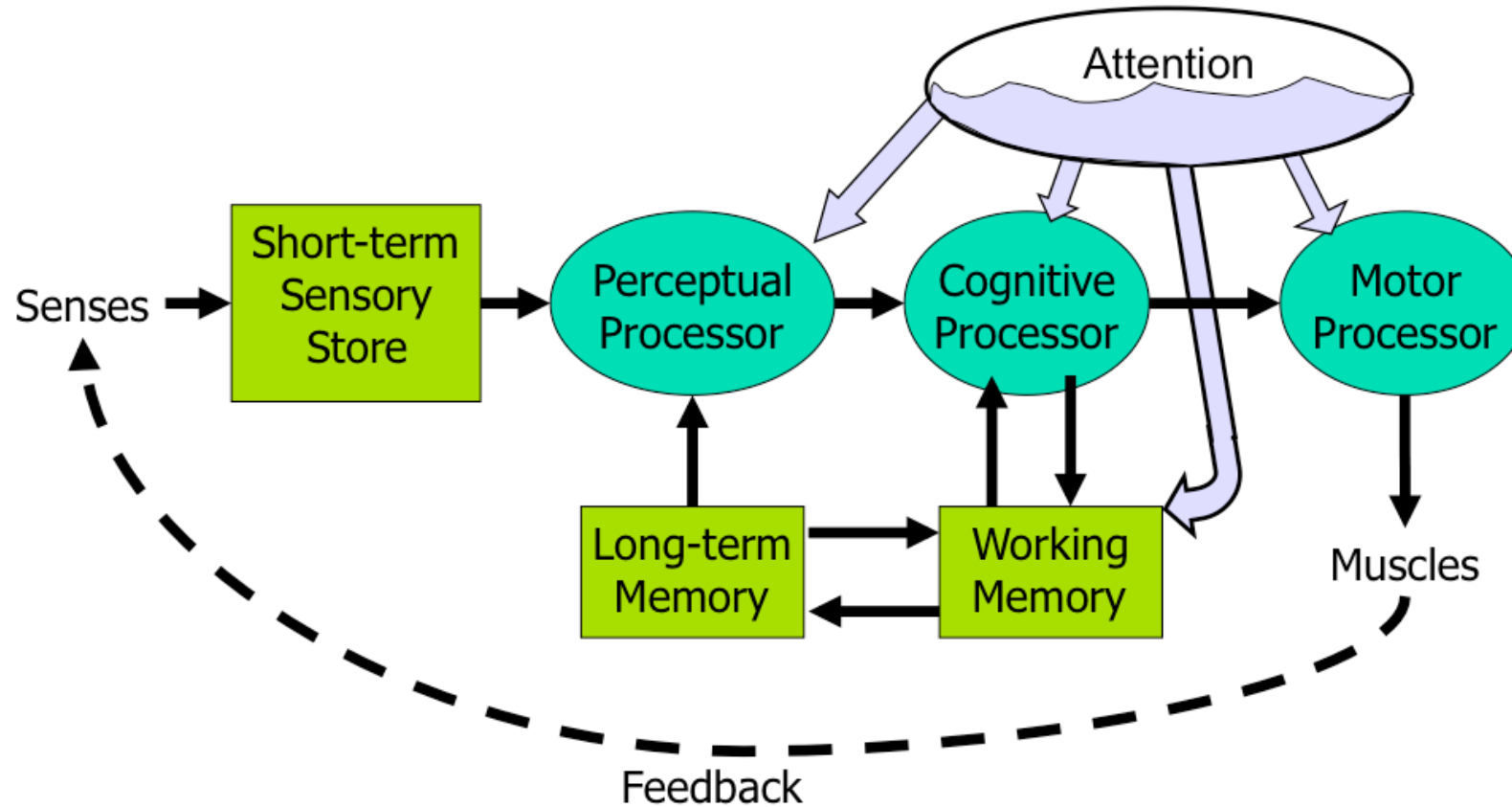
Any Scanner - ...



CGV Cinemas



Human information processing



(Source: MIT CS Course 6.813/6.831)

Mini experiment

- Try to remember items below as many as you can



Mini experiment (cont'd)

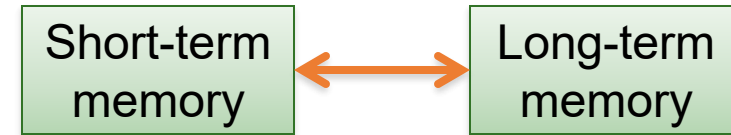
- How many items do you remember?
- How could you remember them?
 - familiar?
 - funny?
 - attracting your attention?
 - related?
 - repeating them?

Memory



- Short-term memory (working memory)

- Small: ~ 7 items or “chunks”
- Short-lived: ~10 seconds
- Repeating helps retain chunks
 - Distraction does the opposite

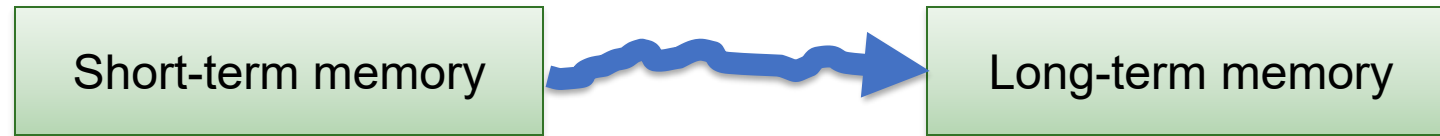


- Long-term memory

- Unlimited size and duration
- Elaborative rehearsal helps transfer chunks from short-term to long-term memory

Learning

- A process of transferring and putting information from short-term to long-term memory



- Implications for user interface design?

Chunking

- Chunk is a unit of memory or perception
 - Depends on how the information is presented

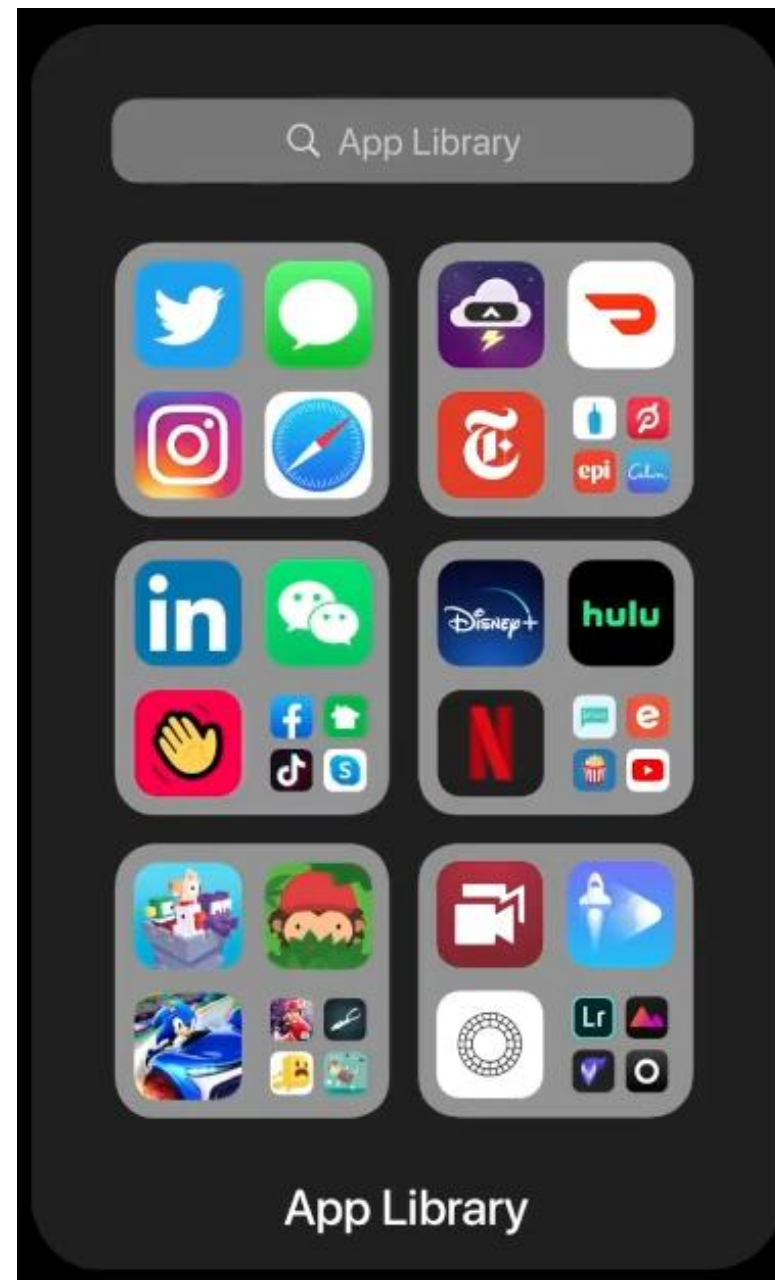
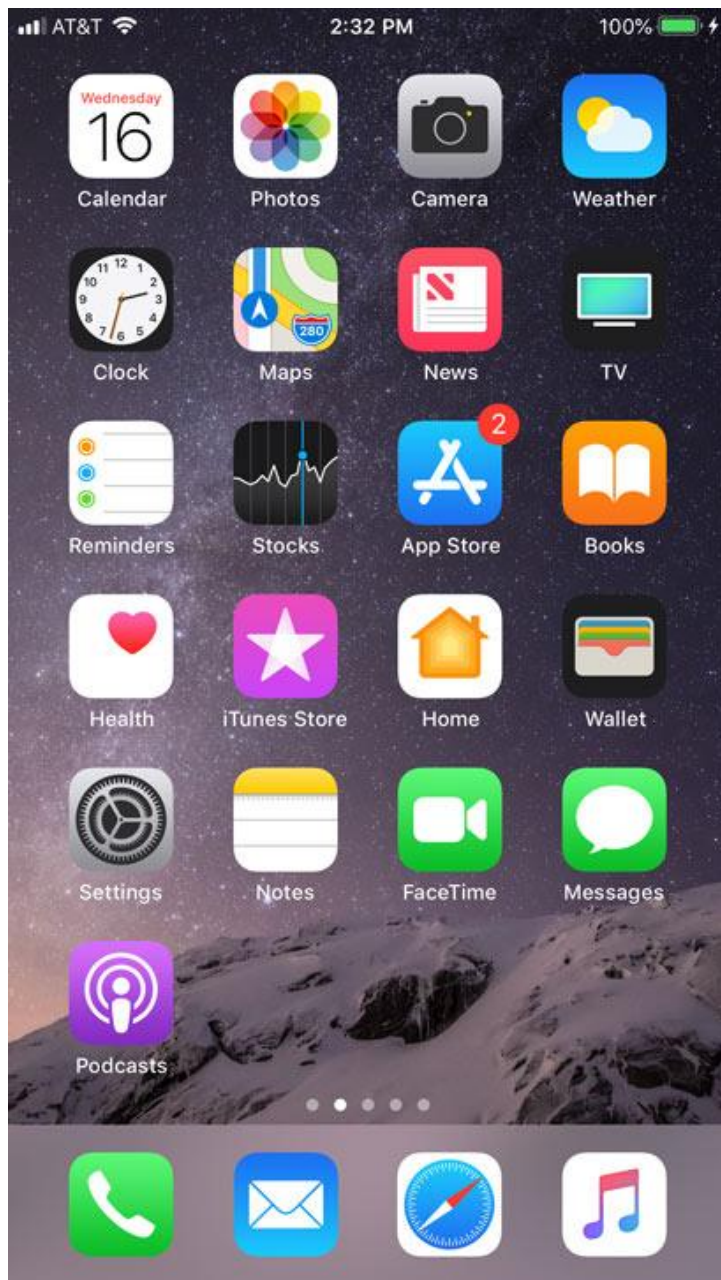
H A PPY V A L E T I N E ← Hard to remember all

HAPPY VALENTINE ← Easy remember all

- Depends on what you already know
 - Linking with the past experience

0888247247 → 0888.247.247

323-481-1585





TRA CỨU TRỰC TUYẾN



Tra cứu mã số BHXH



Tra cứu cơ quan BHXH



Tra cứu CSKCB cấp giấy
nghỉ việc hưởng BHXH



Tra cứu CSKCB ký hợp
đồng khám, chữa bệnh
BHYT



Tra cứu đơn vị tham gia
BHXH



Tra cứu các tổ chức dịch
vụ thu BHXH, BHYT



QL cá nhân



Dịch vụ công



Tra cứu



Trợ giúp



TRA CỨU TRỰC TUYẾN



Tra cứu mã số BHXH



Tra cứu cơ quan BHXH



Tra cứu đơn vị tham gia
BHXH



Tra cứu các tổ chức dịch
vụ thu BHXH, BHYT



Quản lý cá nhân



Dịch vụ công



Tra cứu



Trợ giúp

23:09

57%

Tìm số điện thoại chuyển tiền

99+



Nạp/Rút



Nhận tiền



QR Thanh toán



Quét mã

Quản lý 3 >



Chuyển tiền



Chuyển tiền
Ngân hàng



Thanh toán
hóa đơn



Nạp tiền điện
thoại



Data 3G/4G



Túi Thần Tài



Ví Trả Sau



Thanh toán
khoản vay



Du lịch - Đi lại



Mua vé xem
phim



Heo Đất MoMo



Xem thêm
dịch vụ

MoMo đề xuất



Lắc Xì 2024



Tạo QR bằng
ảnh AI



Quản lý chi
tiêu



Vay Nhanh



MoMo



Ưu đãi



Quét mọi QR



Lịch sử GD



Tôi

23:09

57%

Tìm số điện thoại chuyển tiền



Nạp/Rút



Nhận tiền

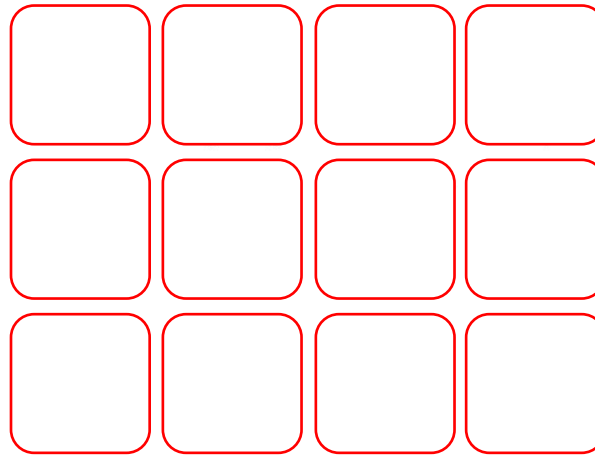


QR Thanh toán



Quét mã

Quản lý 3 >



MoMo đề xuất



Lắc Xì 2024



Tạo QR bằng ảnh AI



Quản lý chi tiêu



Vay Nhanh



MoMo



Ưu đãi



Quét mọi QR



Lịch sử GD

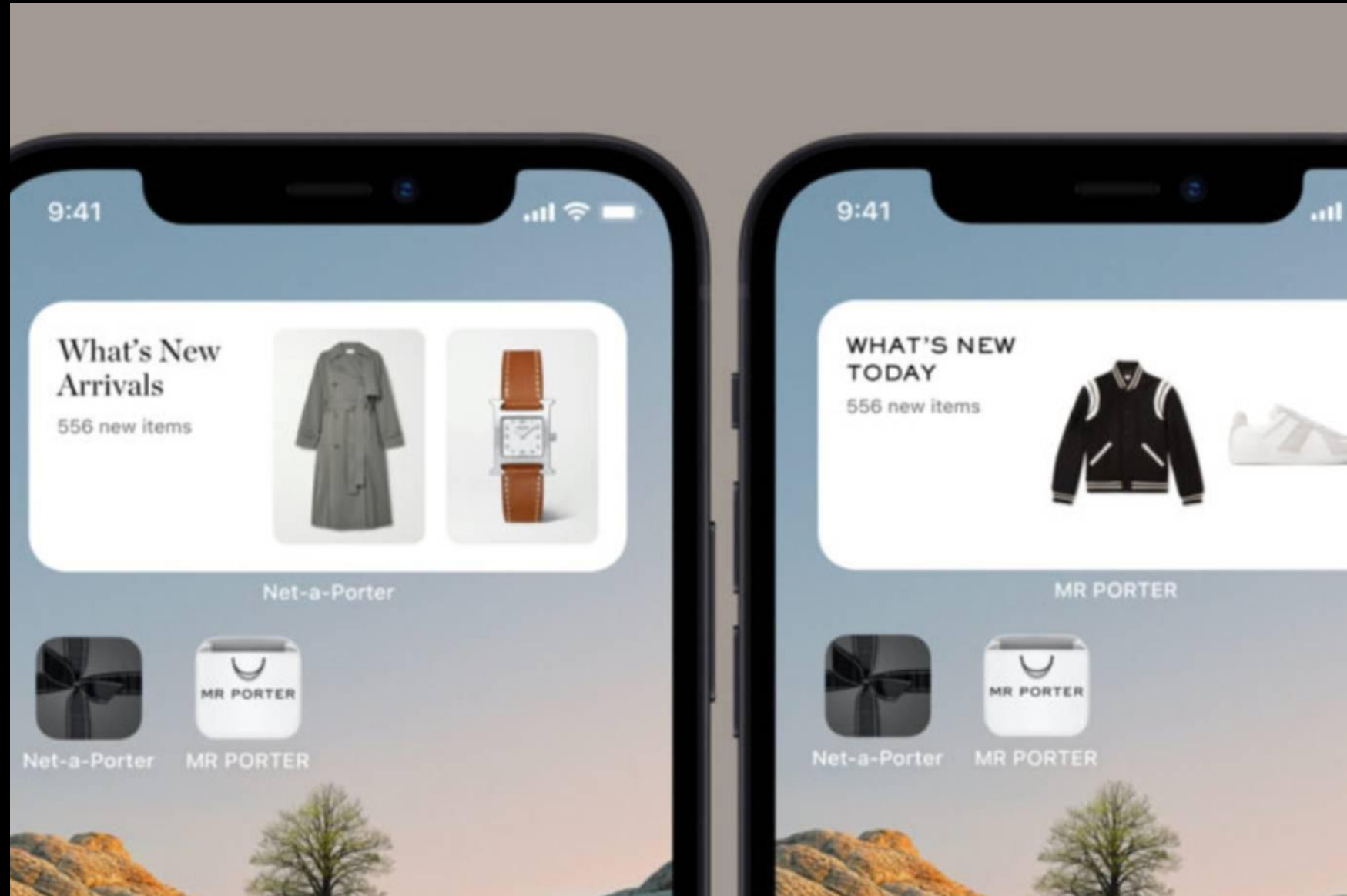


Tôi

iOS Shortcut UIs



Mr.Porter Glanceable UI



Recognition versus recall



Recognition and Recall

- Recognition

- Remembering with the help of a visible cue (evidence – bằng chứng)
- e.g., you recognize your friend easily when seeing his/her face, but you may not remember his/her name.

Một giao diện tốt là giao diện *

- ☐ Dễ học
- ☐ Hiệu quả
- ☐ Dễ nhớ
- ☐ Có nhiều màu sắc
- ☐ Tránh và phục hồi lỗi tốt
- ☐ Mới lạ

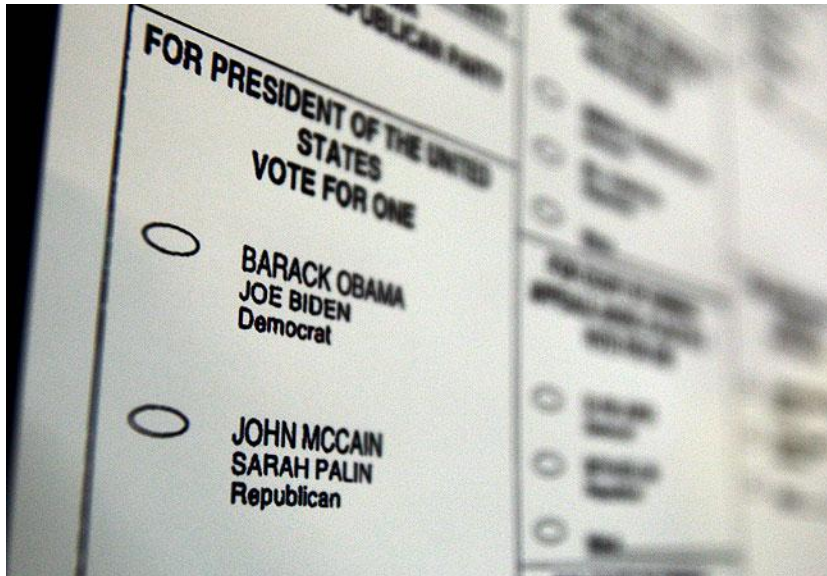
Recognition and Recall (cont'd)

- Recall
 - Remembering with no help
 - e.g., you remember a person when someone refers to him
 - Do you remember her name?
- It is easier to recognize than recall things
 - You don't remember every items in the File menu of Notepad, do you?
 - But you recognize their functions when you look at them



Recognition and Recall (cont'd)

- Which one is recall and recognition?



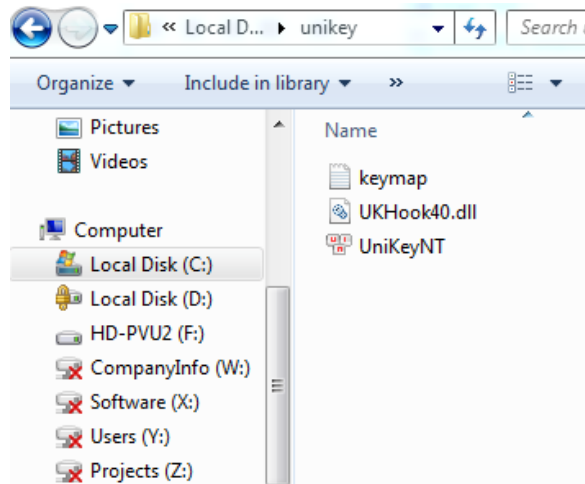
FOR PRESIDENT OF THE UNITED STATES

WRITE NAMES OF THE CANDIDATES TO VOTE FOR

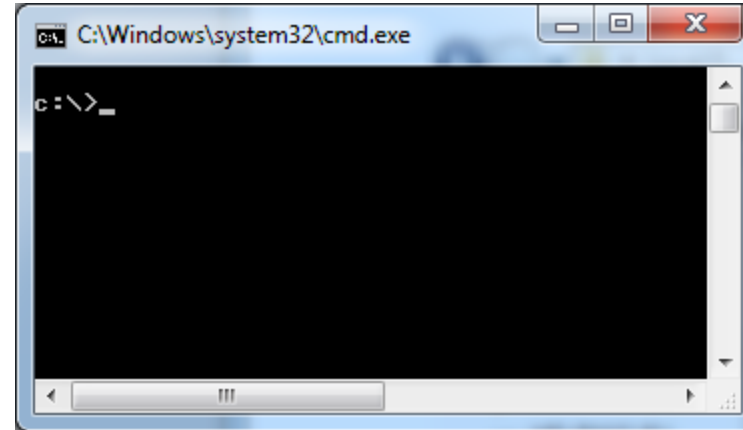
Recognition and Recall (cont'd)

- Implications
 - Performing operations via visual presentation is more learnable than via command line
 - Direct manipulation is more learnable than other interface styles

Delete a file name keymap.txt

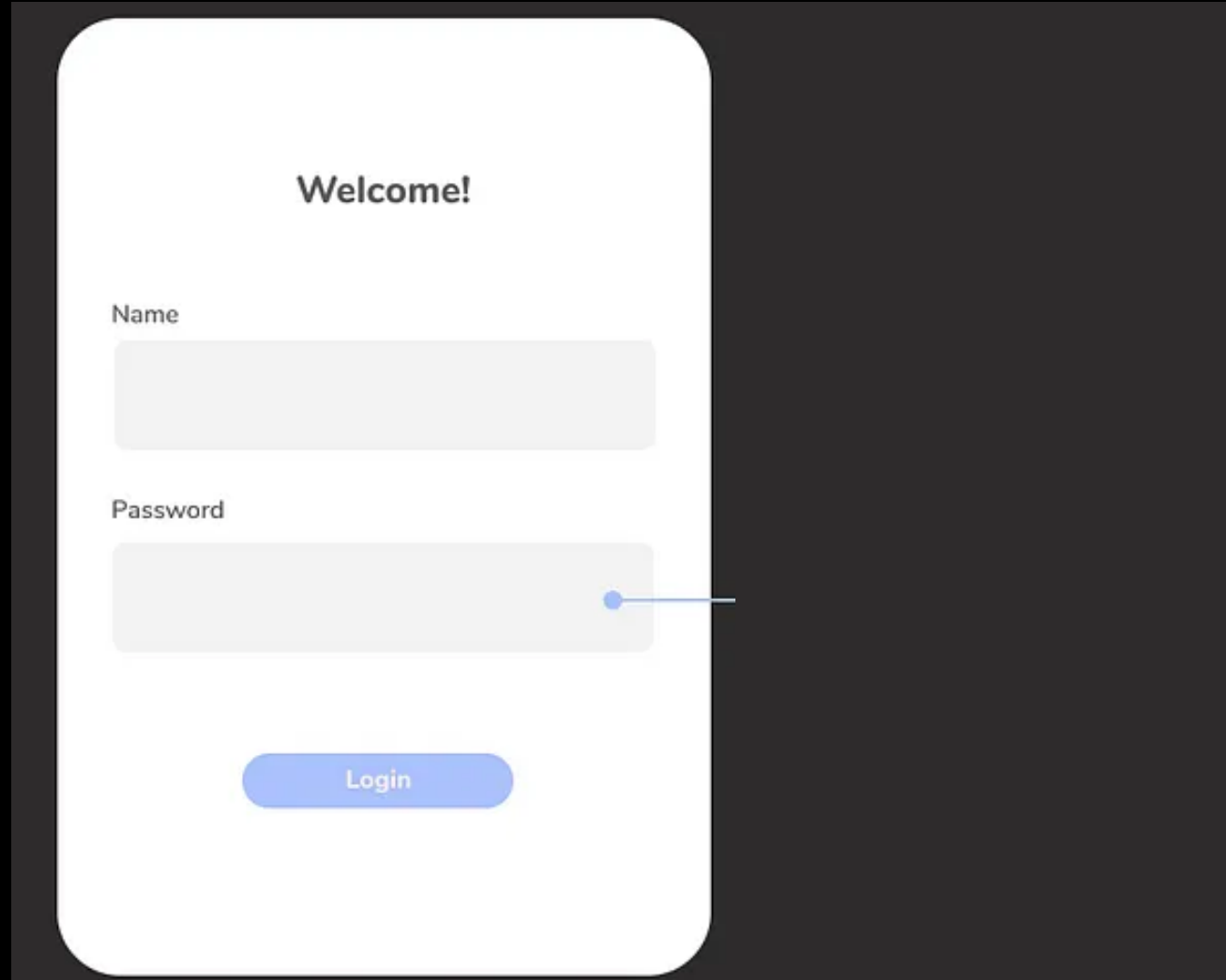


Delete a file name keymap.txt



What do you need to remember to do?

Recall in UI



Welcome!

Name

Password

Login



Welcome

Log in to Auth0 to continue to Auth0
Community.

Continue

OR



Continue with Microsoft Account



Continue with GitHub



Continue with Google

8:40

Cancel



How to Set Up Face ID

First, position your face in the camera frame. Then move your head in a circle to show all the angles of your face.

Get Started

[Accessibility Options](#)



8:40

Cancel



Move your head slowly to complete the circle.



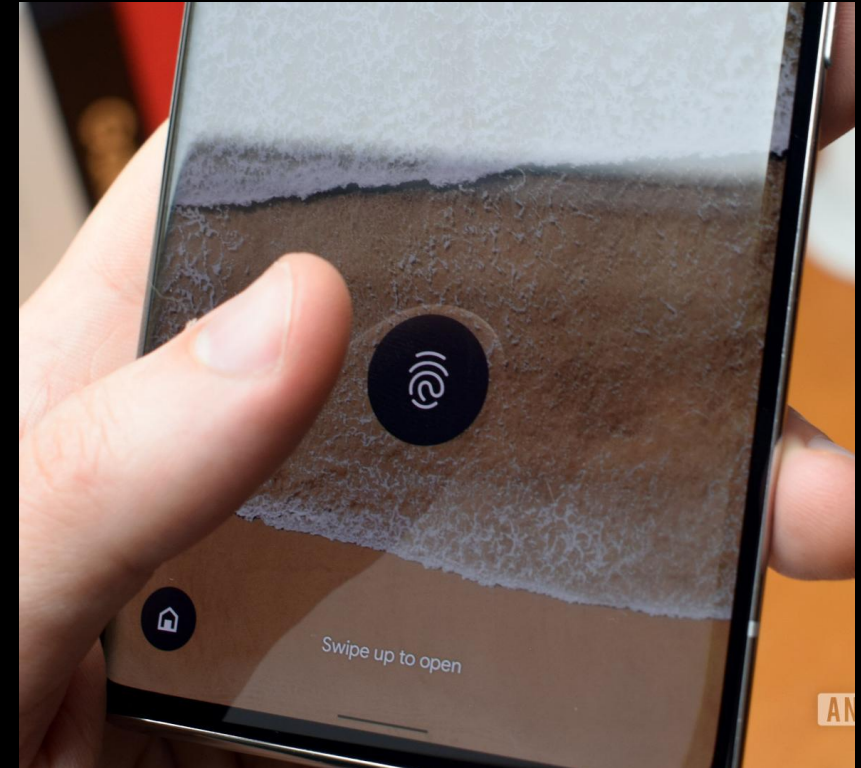
8:40

Cancel



First Face ID scan complete.

Continue



NEW SERIES MONEY HEIST



Play



My List

Watch Part 3 Now

With millions of euros and their lives on the line, nine robbers attempt to pull off the greatest heist of all time.

TV-MA

New Release



NSARI

RIGHT NOW

ON JUNE 12

Emmy Winner • Prison • Gal Pals

WHITNEY
CUMMINGS:
CAN I TALK TO U?CONNERS
IN CARS
GETTING COFFEE

NEW EPISODES



OTHERHOOD

BASKETBALL
THE FIGHTING

Continue Watching

MONEY
HEIST

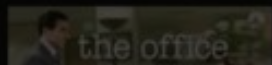
NEW EPISODES

THE GREAT
HACKI AM
MOTHERDONALD GLOVER
WEIRDOBLACK HOLE
APOCALYPSE

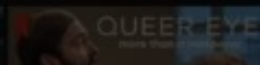
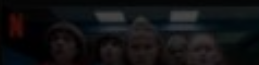
NOW ON

THE MINDY
PROJECTTHE MINDY
PROJECT

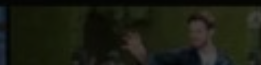
Trending Now



the office



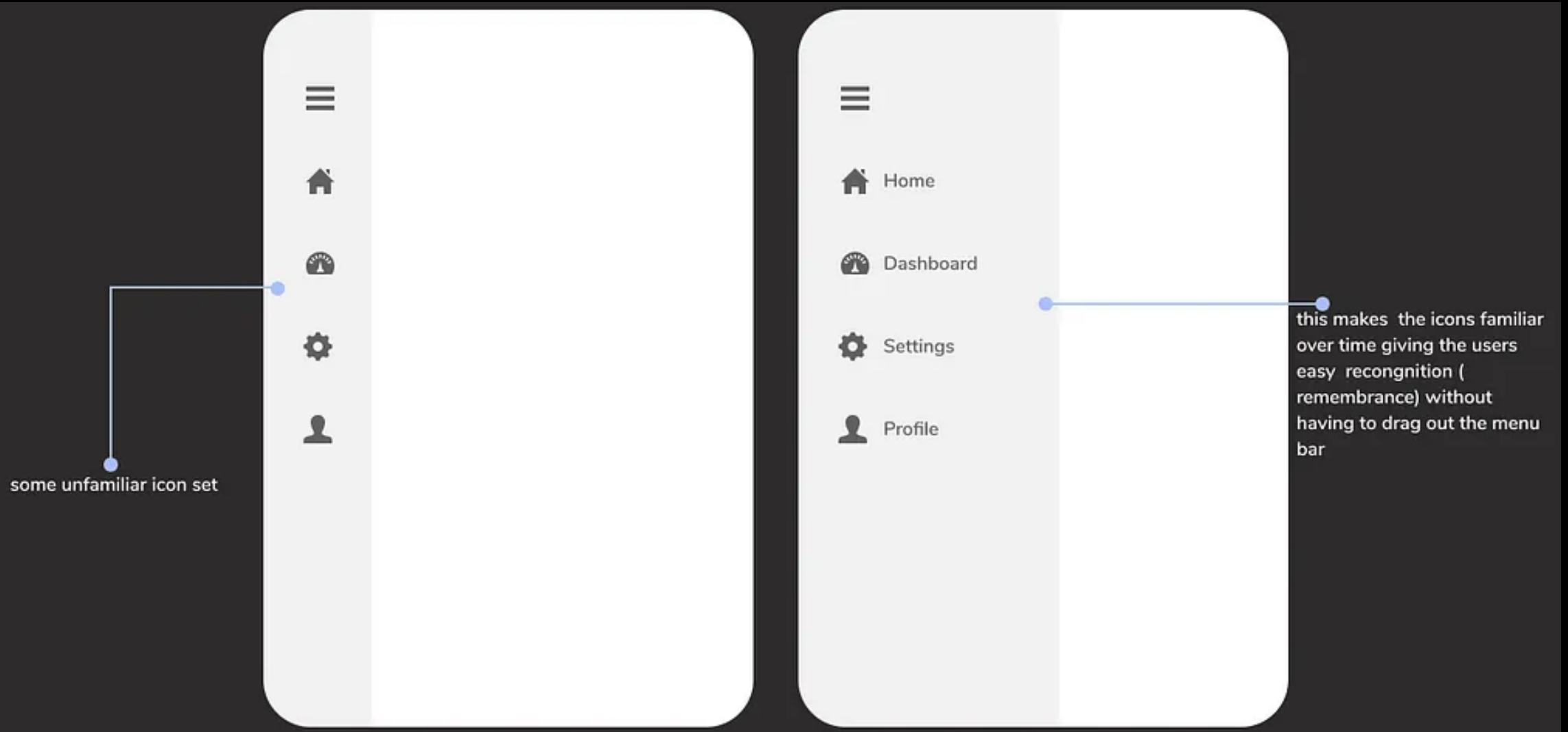
QUEER EYE



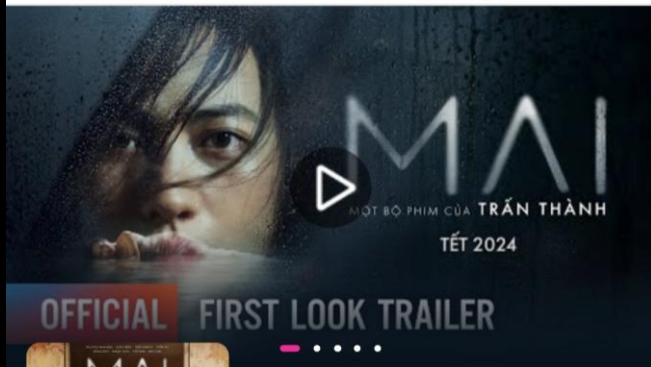
Recognition in UI



Support Recognition in UI



< MAI



MAI

Lãng Mạn

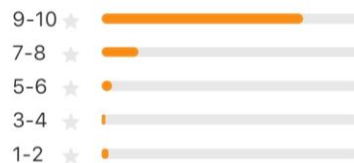
18+ Phim được phổ biến đến người xem từ đủ 18 tuổi trở lên

Ngày khởi chiếu
10/02/2024

Thời lượng
151 phút

Ngôn ngữ
Phụ đề

★ **9.1** /10
(30.5k đánh giá)



Đáng xem (9.6k)

Cảm động (6.4k)

Tuyệt vời (4.5k)

Hài lòng (4.2k)

Mua vé

MAI

MAI

MỘT BỘ PHIM CỦA TRẦN THÀNH

TẾT 2024

OFFICIAL FIRST LOOK TRAILER

MAI

LĂNG MẠN

18+

Phim được phổ biến đến người xem từ đủ 18 tuổi trở lên

Ngày khởi chiếu

10/02/2024

Thời lượng

151 phút

Ngôn ngữ

Phụ đề

★9.1/10

(30.5k đánh giá)

Đáng xem (9.6k)

Cảm động (6.4k)

Tuyệt vời (4.5k)

Hài lòng (4.2k)

Mua vé

MAI

89k/vé

CẢ TUẦN

mo mo LOTTE CINEMA

3/3

H.nay

Thứ 3

Thứ 4

Thứ 5

Thứ 6

Thứ 7

C.N

26

27

28

29

01

02

0

Tất cả

9:00 - 12:00

12:00 - 15:00

15:00 - 18:00

Đề xuất

CGV

Lotte

Galaxy

Rạp đề xuất (5)

TP.HCM

LOTTE CINEMA

Lotte Nowzone

Bạn ở gần rạp này • 0.2 km

Tầng 5, TTTM Nowzone, 235 Nguyễn Văn... [Tìm đường](#)

2D Lồng tiếng

12:15 ~ 14:46

138/138 Ghế

14:00 ~ 16:31

136/138 Ghế

14:45 ~ 17:16

136/138 Ghế

15:30 ~ 18:01

103/103 Ghế

16:30 ~ 19:01

138/138 Ghế

17:15 ~ 19:46

130/138 Ghế

18:00 ~ 20:31

103/103 Ghế

19:00 ~ 21:31

138/138 Ghế

19:45 ~ 22:16

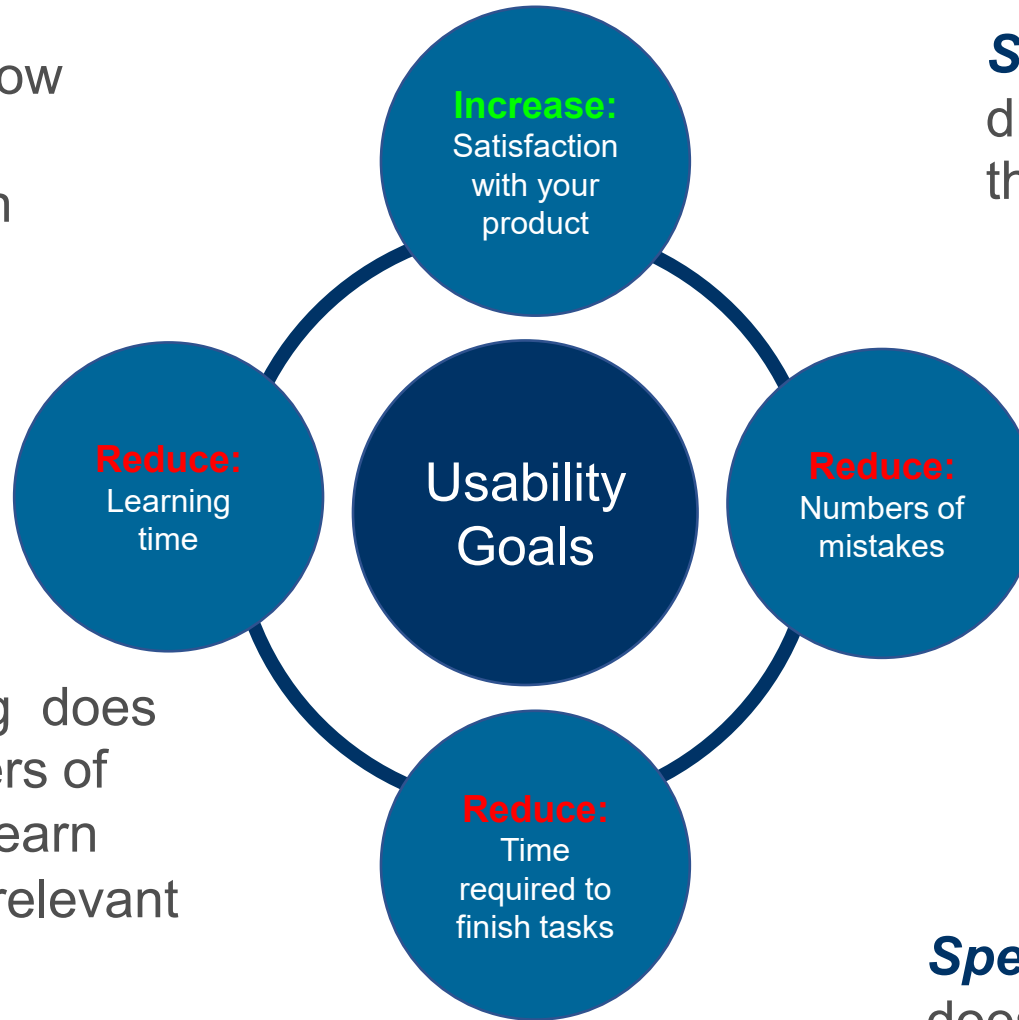
130/138 Ghế

Usability definition

- Usability
 - How well users can use the system's functionality
- Dimensions
 - Learnability
 - How easy it is to learn and use?
 - Efficiency
 - How quickly users perform tasks using the UI?
 - Memorability
 - How easy it is for users to reestablish proficiency?
 - Errors
 - Are the errors committed by users often? Is it easy to recover from errors?
 - Satisfaction
 - Are users satisfied with the UI?

USABILITY GOALS AND MEASURES

Retention over time: How well do users maintain their knowledge after an hour, a day, or a week?



Subjective satisfaction: How much did users like using various aspects of the interface?

Rate of errors by users: How many and what kinds of errors do people make in carrying out the benchmark tasks?

Time to learn: How long does it take for typical numbers of the user community to learn how to use the actions relevant to a set of tasks?

Speed of performance: How long does it take to carry out the benchmark tasks?

Remember this?

UX vs. Usability

Evaluation of understandability vs. holistic design

Designing it right vs. the right design

Often, poor usability leads to a bad user experience, but not necessarily

USABILITY EVALUATIONS



HOW TO DESIGN FOR A GOOD USABILITY?

- Visibility
- Accessibility
- Consistency
- Affordances

Visibility

- Operations should be visible to users
 - Unix commands are very invisible vs. Windows' menus
 - Right click menus are not very visible
 - A reason why iOS does not support much right-click
 - Drag-drop is not either
 - But it's a direct manipulation style reflecting real world
- Visibility versus Simplicity
 - More visibility may result in reduced simplicity



VISIBILITY



VISIBILITY



Accessibility

Why is Accessibility Important?

Types of impairments that affect how people use your digital product



VISUAL



AUDITORY



COGNITIVE



MOBILITY

Accessibility



Accessibility



AFFORDANCES

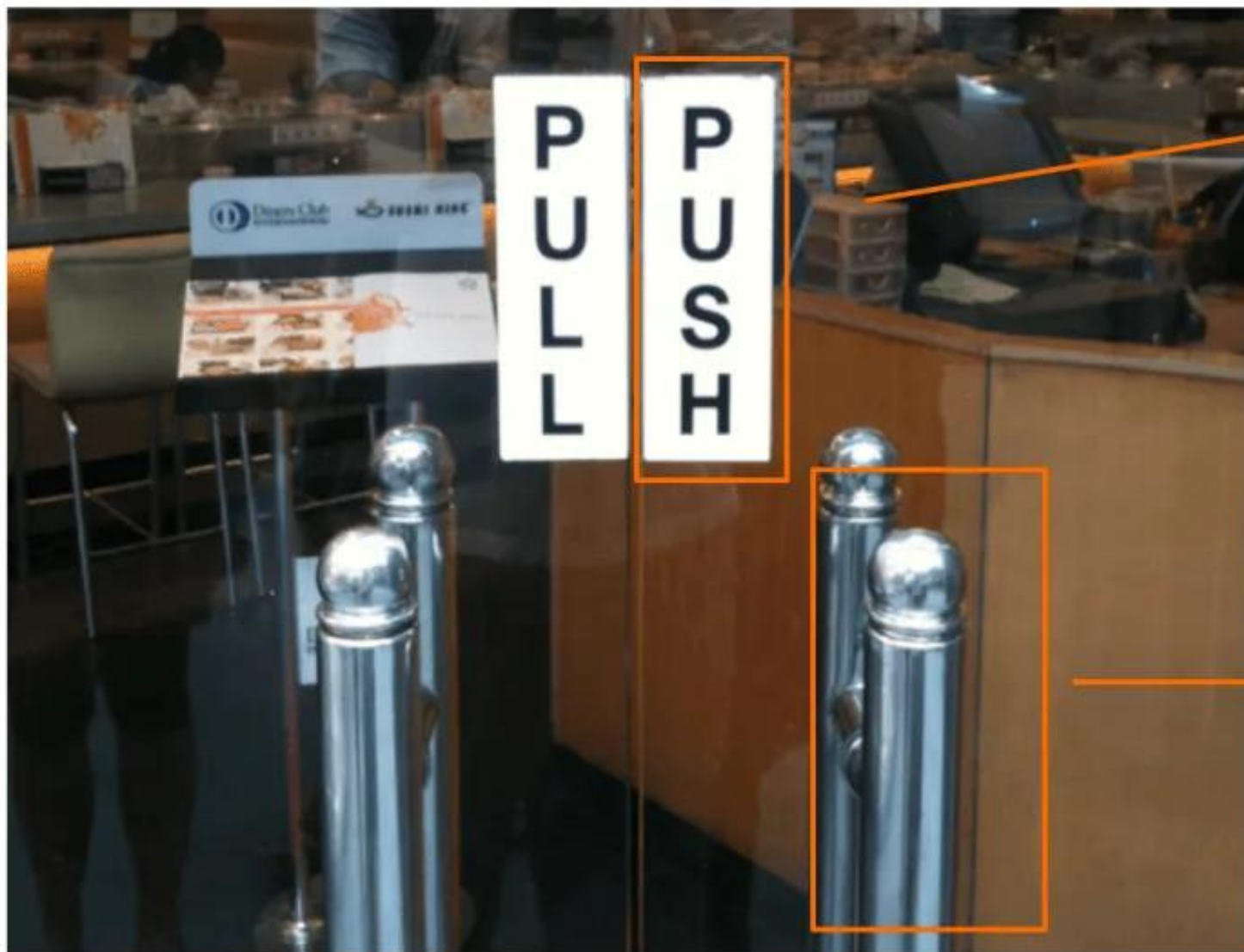
Affordance

- Perceived and actual properties of an object that give clues to its operation



Examples of Affordances





Signifier

- Sign to explain what to do (because the affordances are confusing)

Affordance

- Handle to physically grasp

Which one is a button?



IPSUM

IPSUM

IPSUM

NEW SERIES MONEY HEIST



Play



My List

Watch Part 3 Now

With millions of euros and their lives on the line, nine robbers attempt to pull off the greatest heist of all time.

TV-MA

New Release



NS-RI

RIGHT NOW

ON JUNE 12

[Emmy Winner](#) [Prison](#) [Gal Pals](#)**WHITNEY
CUMMINGS:
CAN I TOUCH IT?****COMEDIANS
IN CARS
GETTING COFFEE**

NEW EPISODES

**OTHERHOOD****BASKETBALL**

CAN NOTHING

Continue Watching

**MONEY
HEIST**

NEW EPISODES

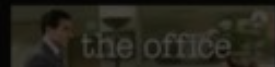
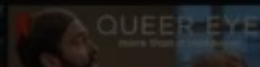
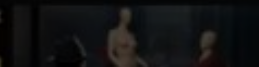
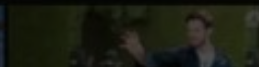
**THE GREAT
HACK****I AM MOTHER****DONALD GLOVER
WEIRDO****BLACK HOLE
APOCALYPSE**

HOMES OFFER

**THE MINDY
PROJECT**

HOMES OFFER

Trending Now

**the office****QUEER EYE**

Natural mapping

- Physical arrangement of controls matches arrangement of their operations
- It's best to map directly, but not always have to be
 - Light switches
 - Car's turn signals



Feedback

- Actions should have immediate effects
 - e.g., push buttons, scroll bars, mouse icons
- Feedback types
 - Audio
 - Visual
 - Haptic (giving a feeling, e.g., vibration of a mouse click)

Consistency

- Similar things should work similarly
 - Fonts, colors, icons, layouts, etc.
- Different things should look different
- Consistency types
 - Internal: within the system
 - External: across different systems
 - Metaphorical: reflecting real-world objects
 - A print icon is a metaphor of the printer

👋 **Bad**

This is an
amazing title

Since company's name isn't
so self-explanatory, landing
section will have clear title
of couple words that will...

This is an
amazing title

Since company's name isn't
so self-explanatory, landing
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**This is an
amazing title**

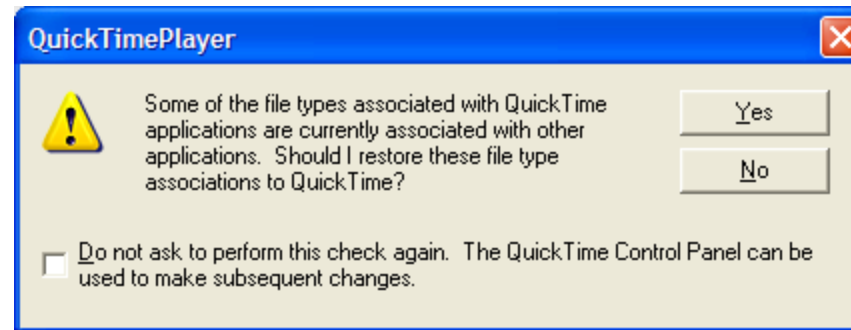
Since company's name isn't
so self-explanatory, landing
section will have clear title of
couple words that will...

**This is an
amazing title**

Since company's name isn't
so self-explanatory, landing
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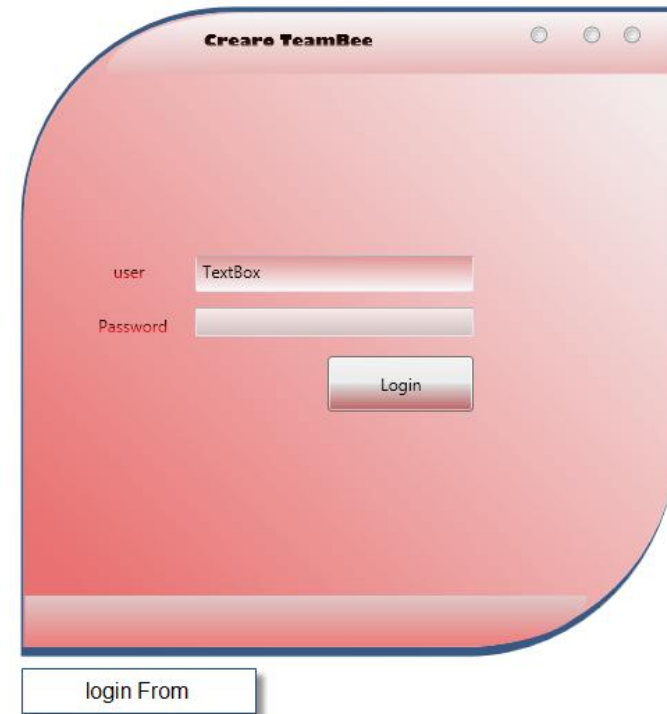
Concistency (cont'd)

- Speak the user's language
 - Use common words, avoid slangs and jargon
 - But avoid wordy and overly verbose



Platform standards

- Follow guidelines of platforms
 - MS Windows user interface guidelines
 - Apple user experience guidelines
- Follow frameworks
 - Various frameworks have their own looks and feels guidelines
- Learn from existing applications

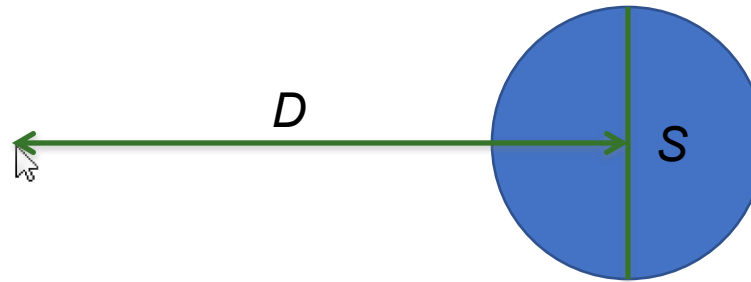


Efficiency

Fitts's law

- Time T to move hand to a target of size S at distance D away from the mouse pointer is

$$T = a + b * \log (D/S + 1)$$



- a and b are constants
- T is dependent only on $\log (D/S + 1)$
- $\log (D/S + 1)$ is defined as *index of difficulty*

Implications of Fitts's law

- Similar targets should be grouped
- Targets at screen edge are easy to hit
- Pie menu is faster to use than linear menu
 - It's faster 15-20% according to a study by Callahan, 1994
- Lengthy menus should be avoided

(Callahan et al. 1994, "An empirical comparison of pie vs. linear menus," CHI 1991)

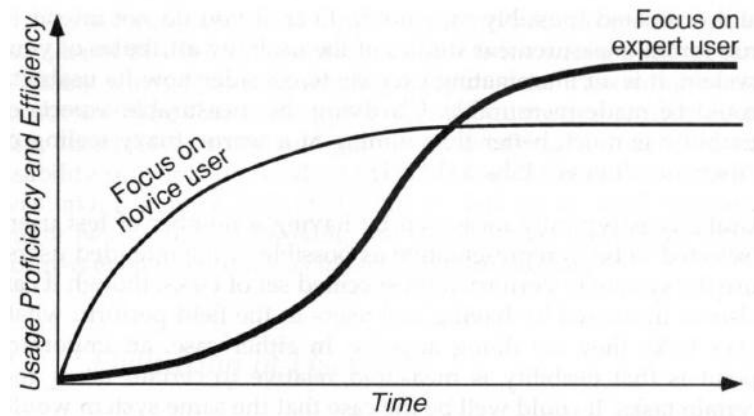
Power law of practice

- Time T_n to do a task the n th time is

$$T_n = T_1 * n^{-a}$$

a is typically from 0.2 to 0.6

- Implications
 - With practice, novices get better
 - But their performance becomes nearly flat
 - Remember the Nielsen's Learning curve?

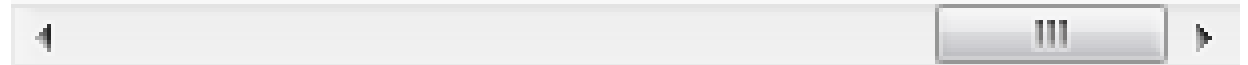


SCROLL BARS & NAVIGATION ARROWS

OSX Snow Leopard



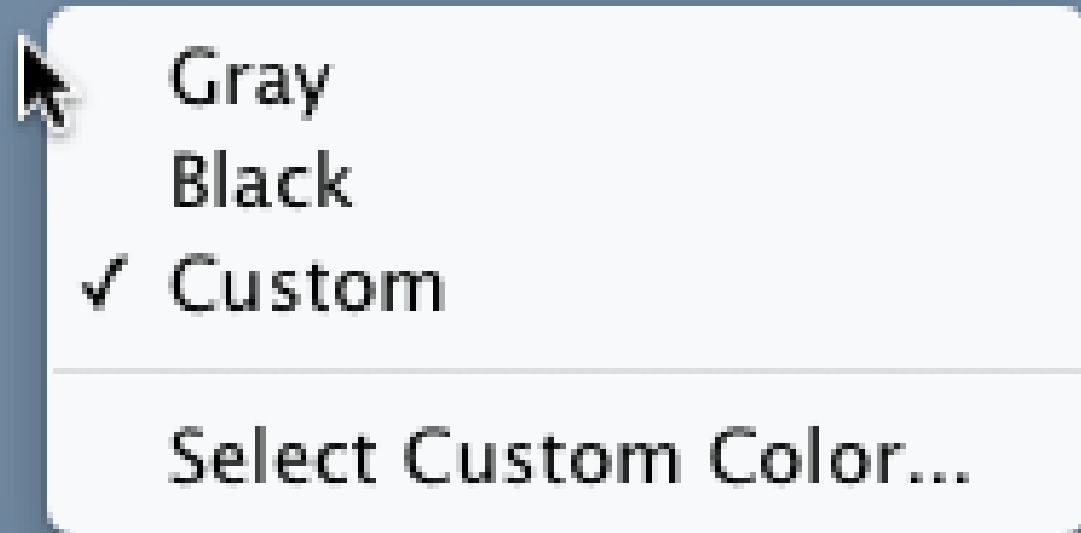
Windows



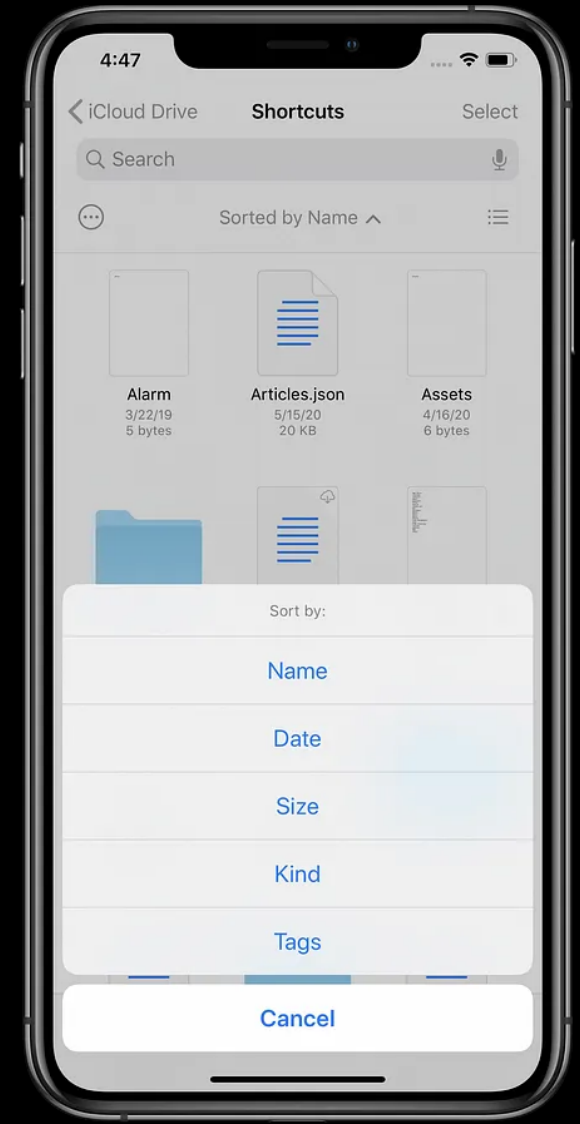
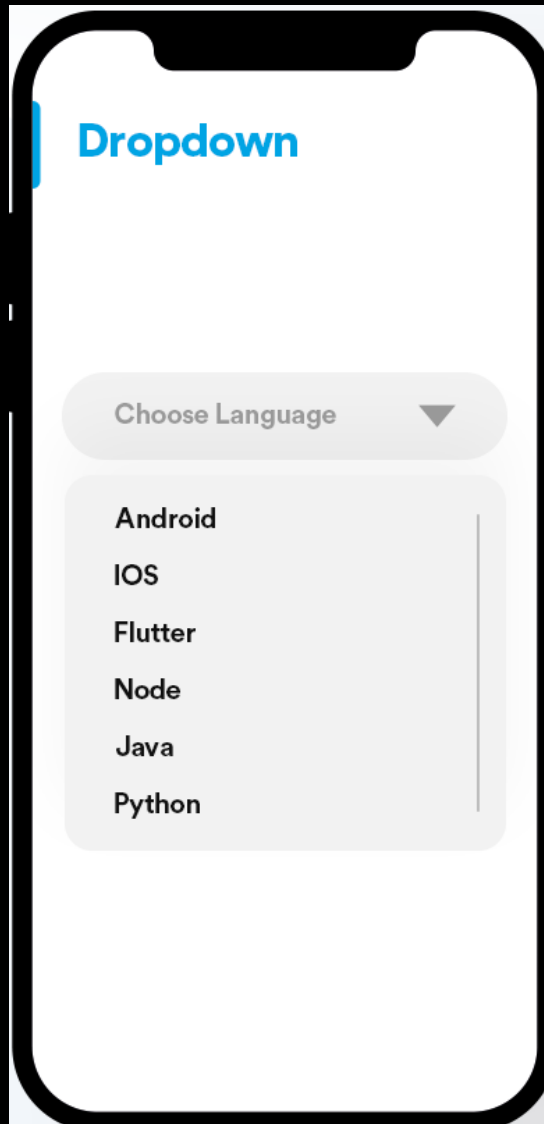
Gmail Threw Discard Button Away

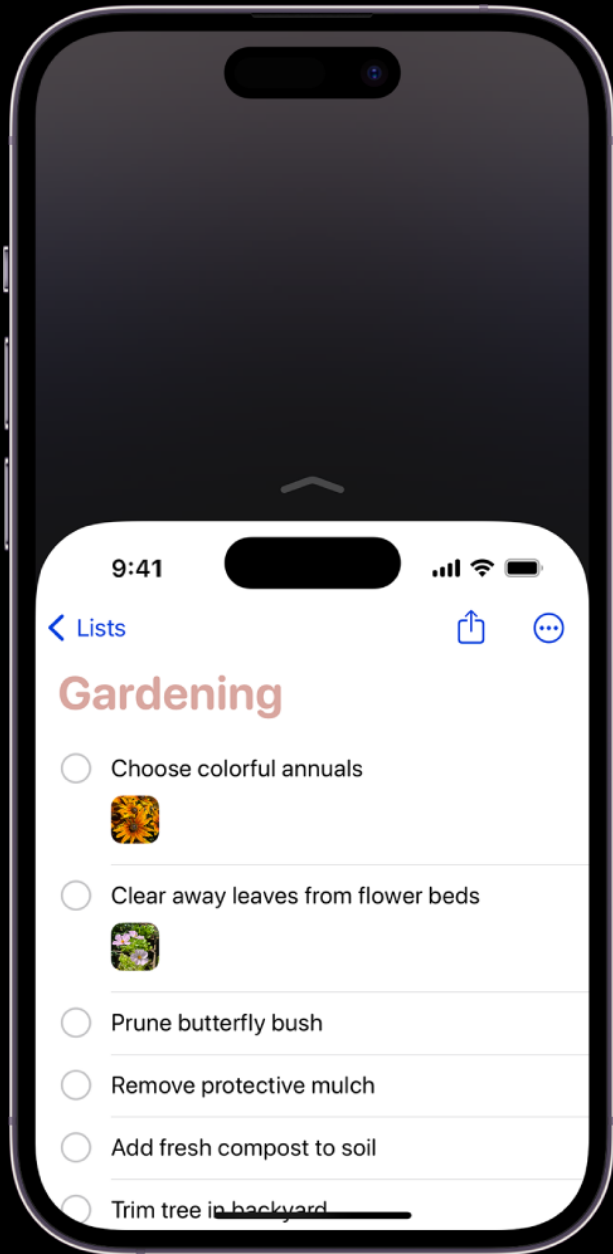


POP-UP MENUS



HISTORY OF MOBILE DROPDOWN LIST UI





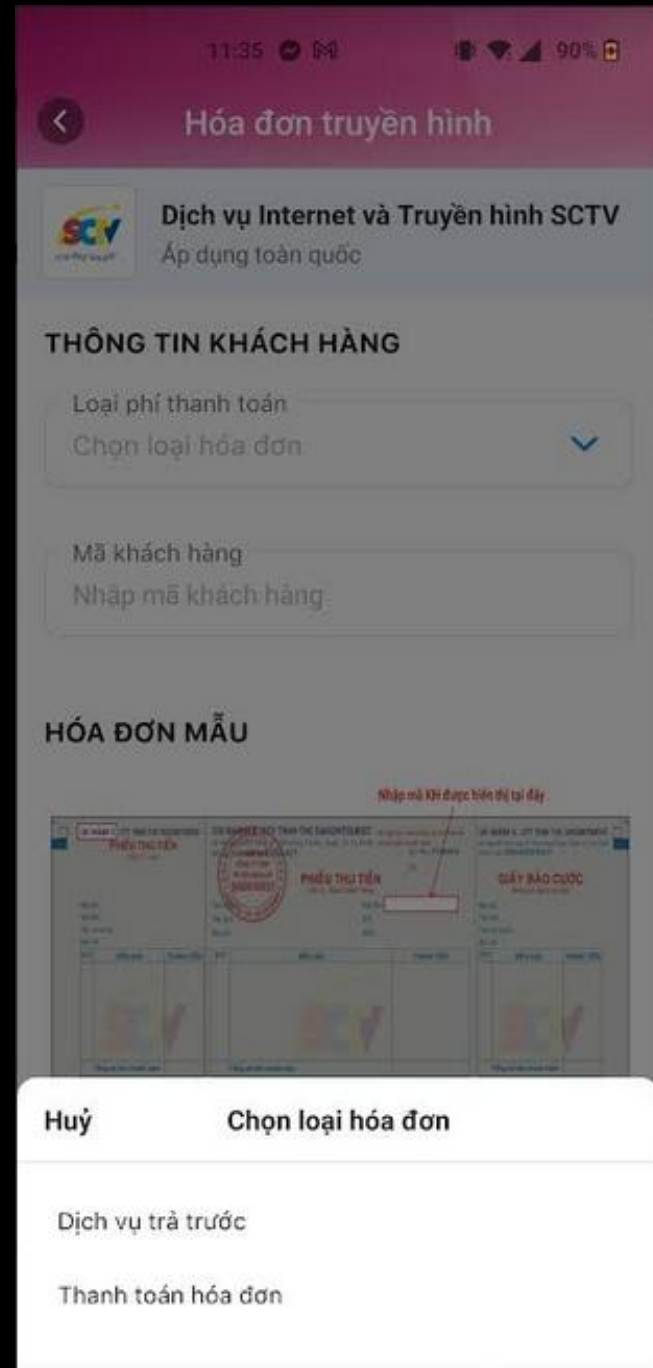
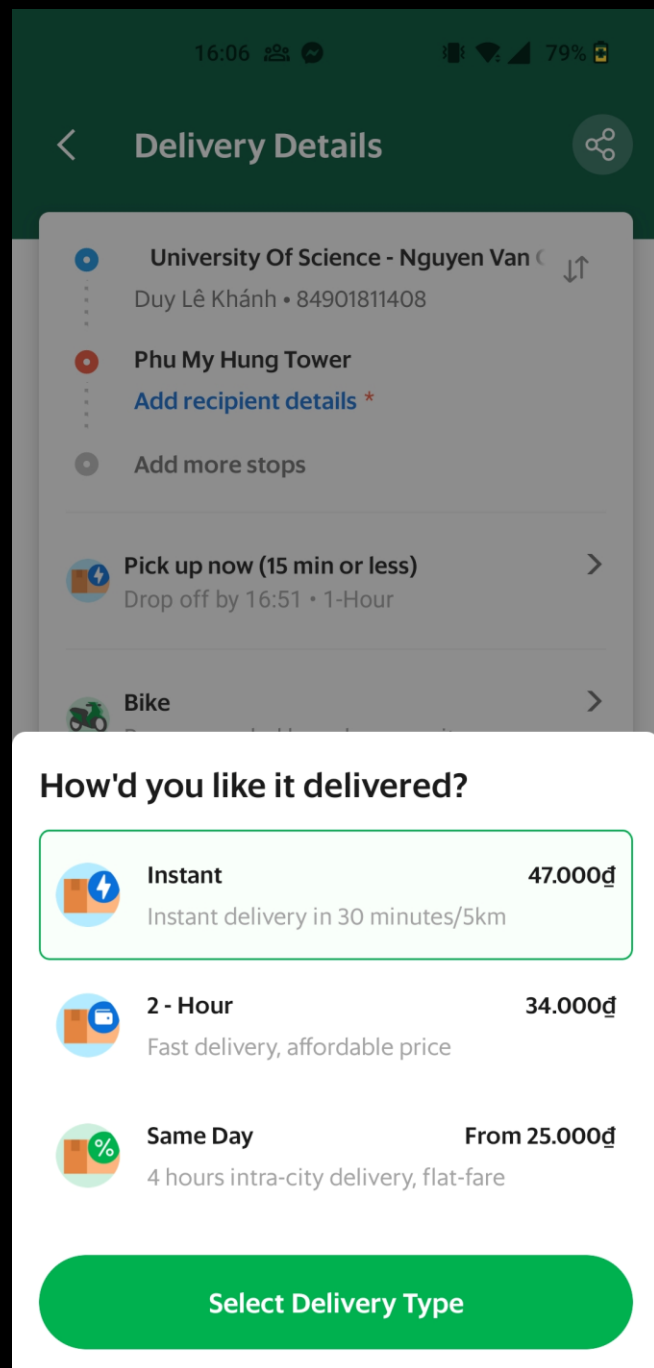
1. Go to Settings  > Accessibility > Touch, then turn on Reachability.

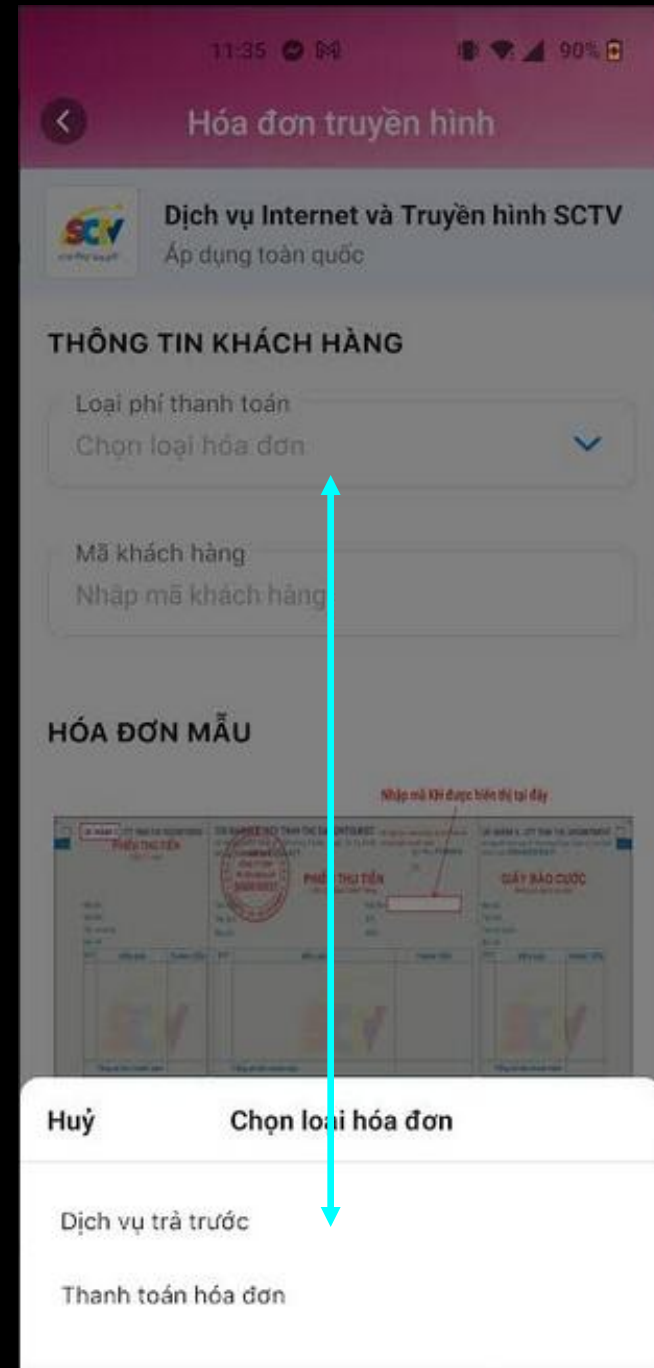
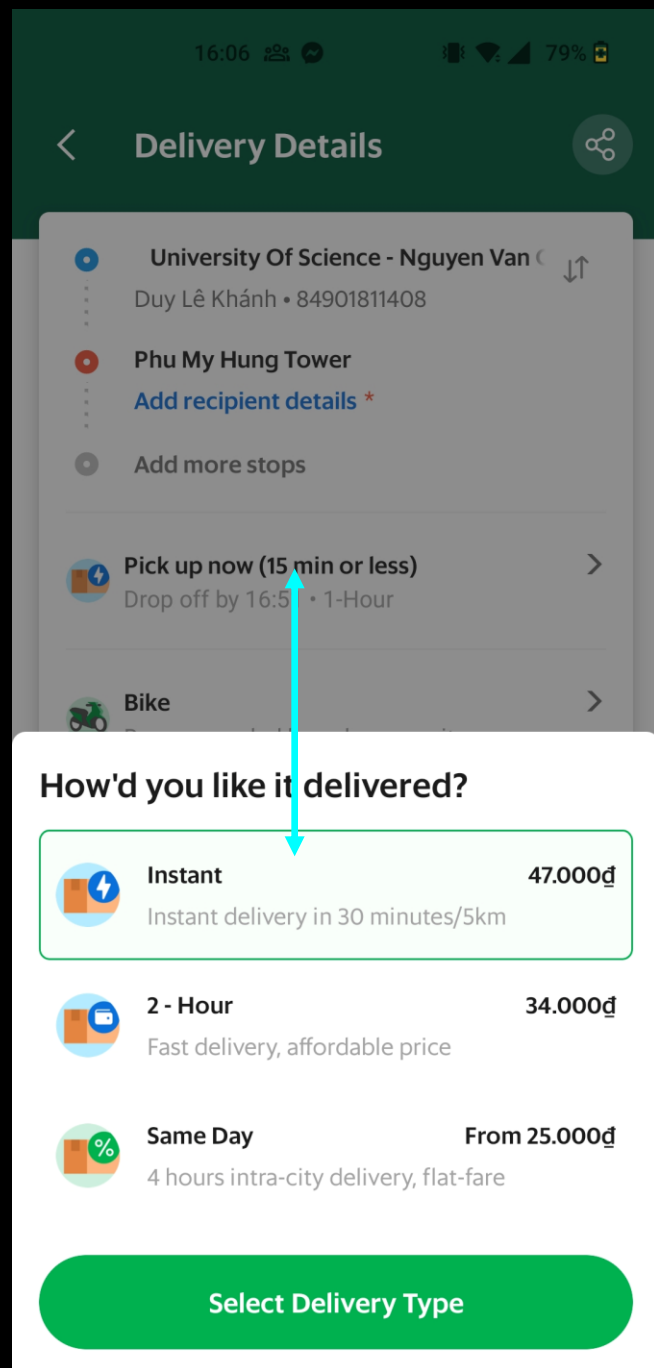
2. To lower the top half of the screen, do one the following:

- *On an iPhone with Face ID:* Swipe down on the bottom edge of the screen.
- *On an iPhone with a Home button:* Lightly double-tap the Home button.

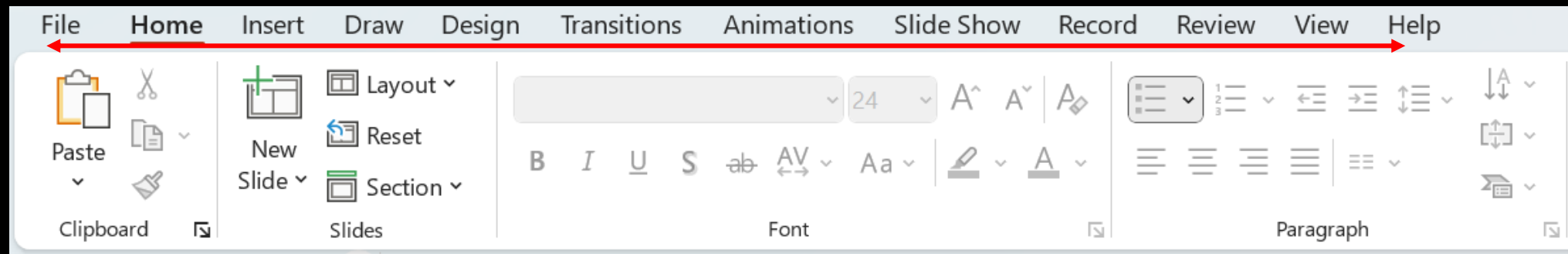
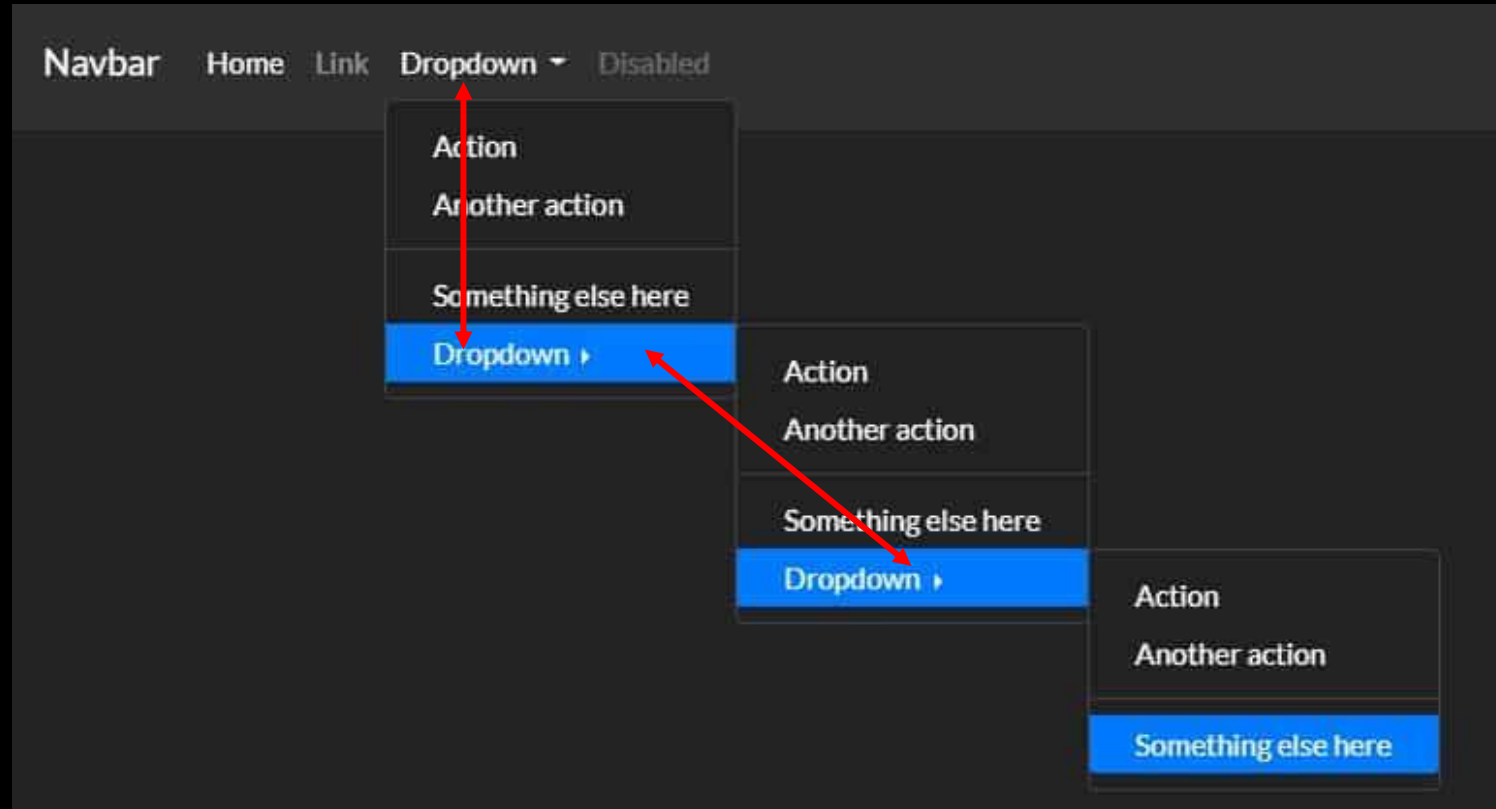
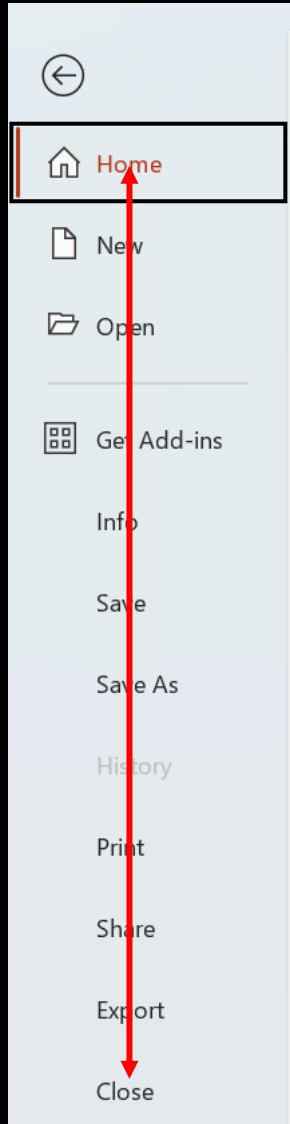
3. To return to the full screen, tap the upper half of the screen.

<https://support.apple.com/en-vn/guide/iphone/iph145eba8e9/ios>

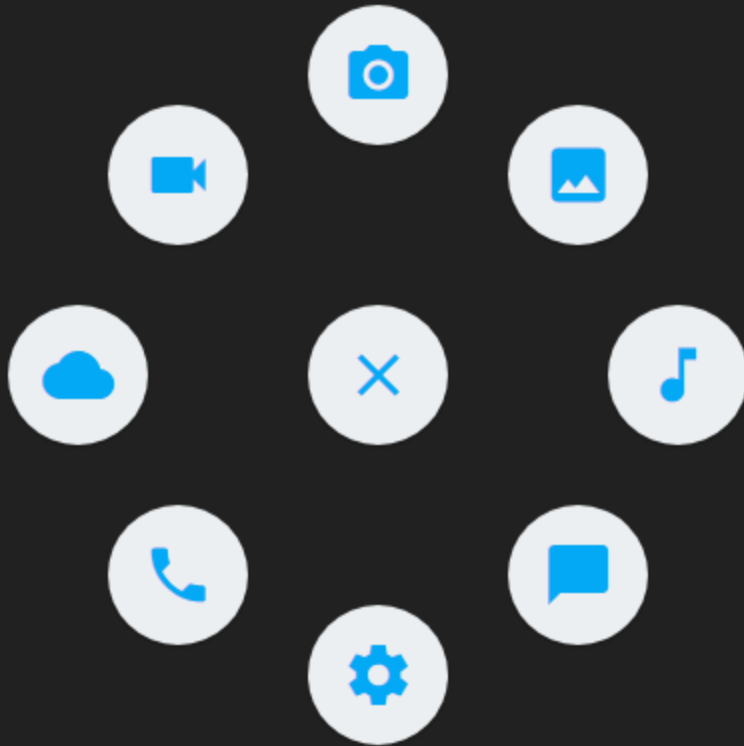




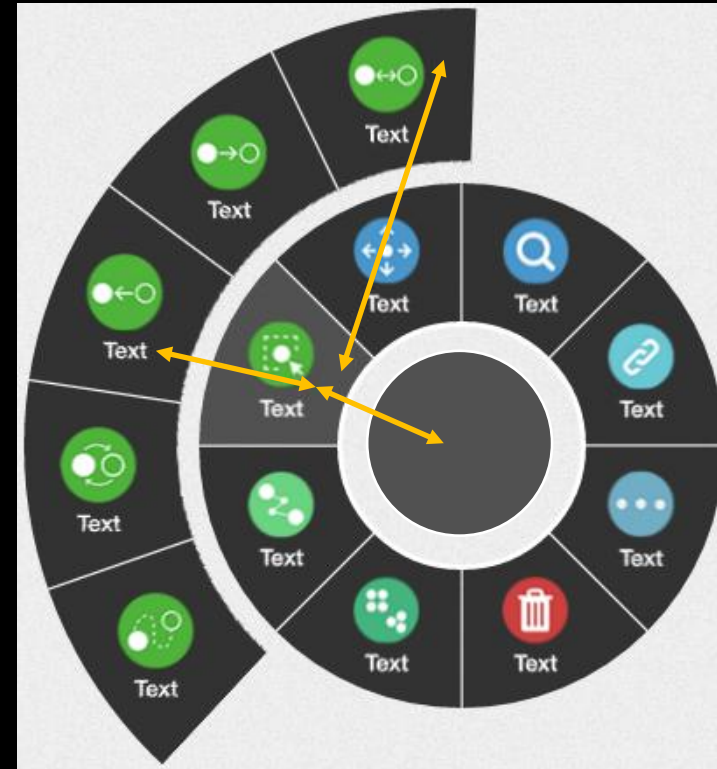
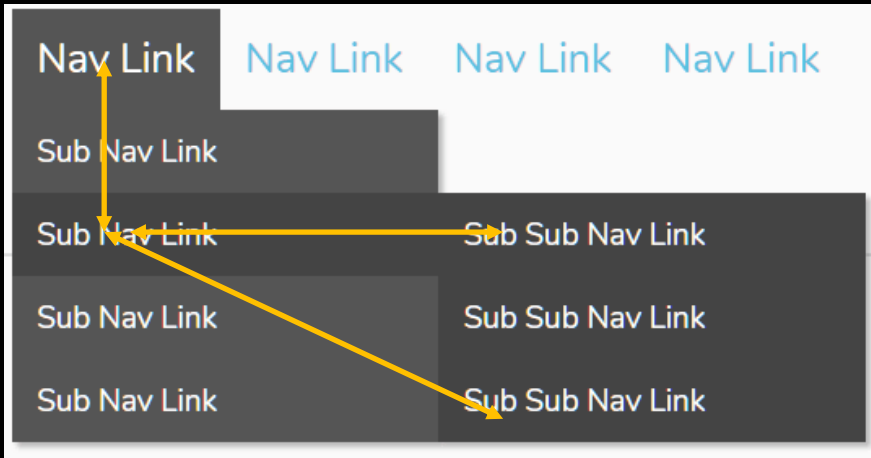
Problem with list-based menu



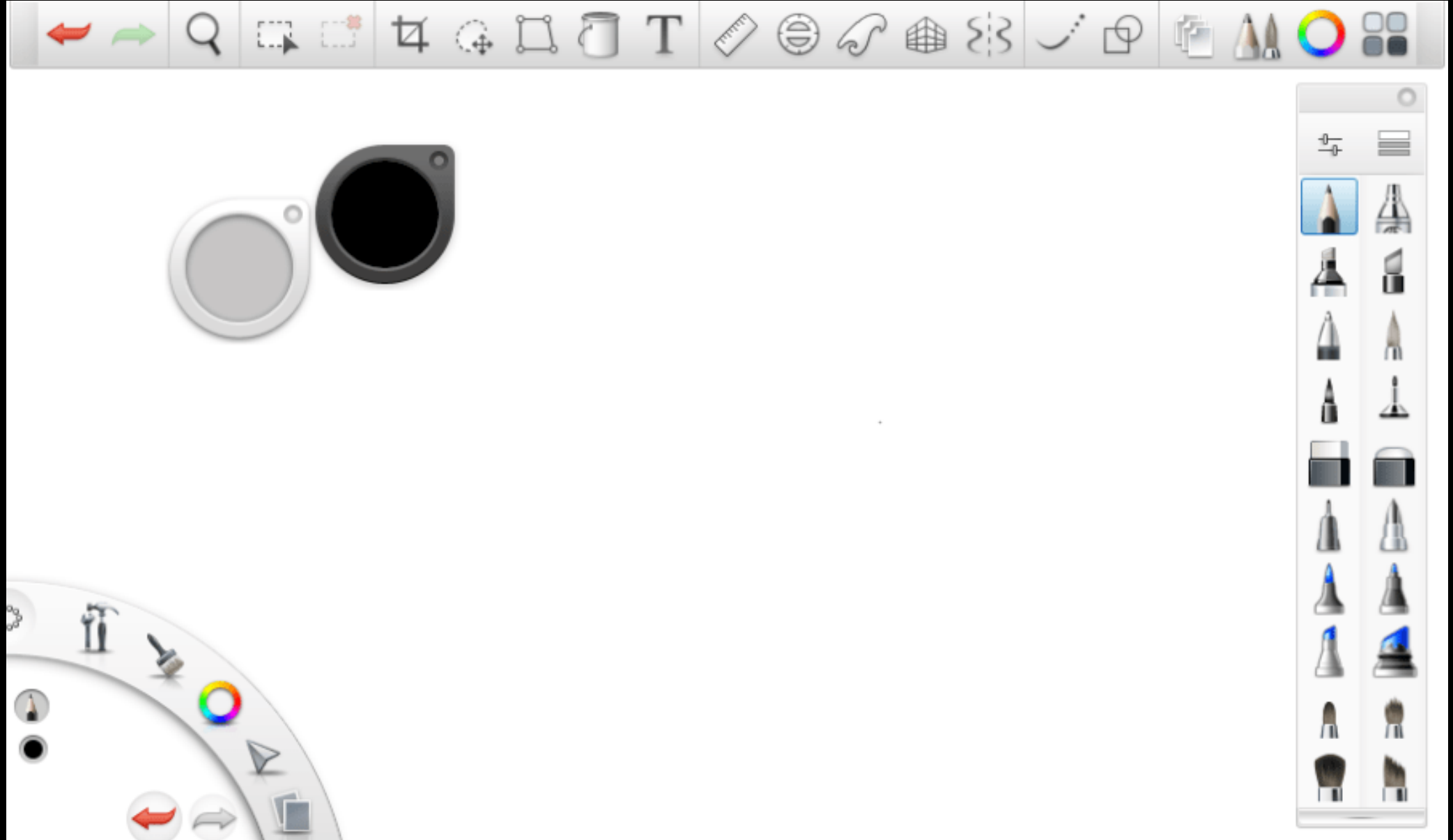
CIRCULAR MENU



- Shorter average acquisition time
- Support spatial memory



AUTODESK SKETCHBOOK

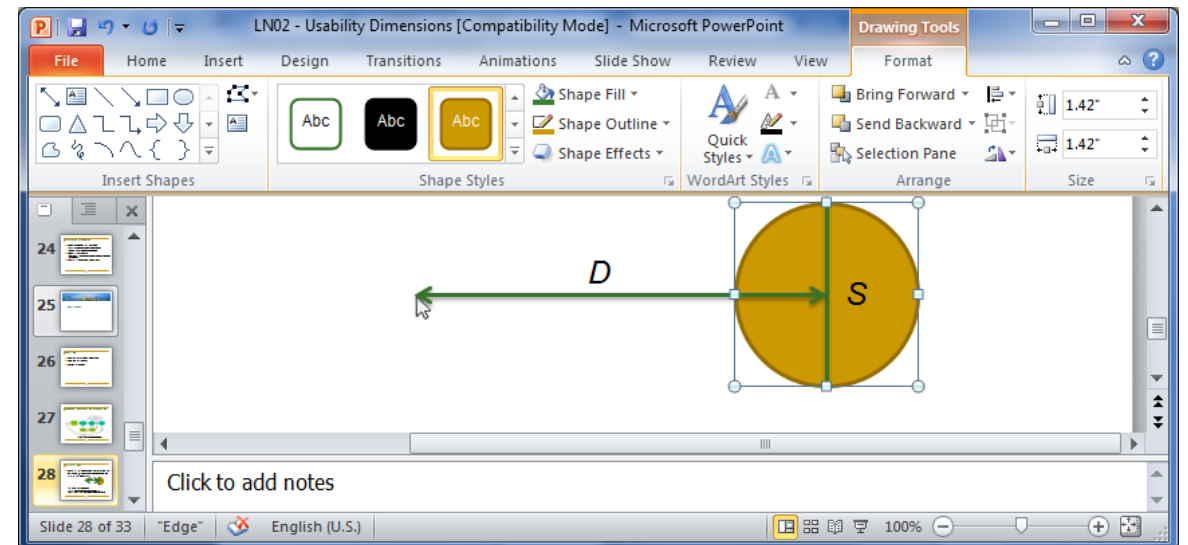


NOKIA (LUMIA) CAMERA APP



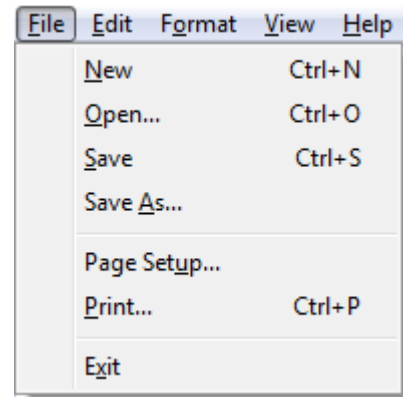
Principles to improve efficiency

- Make often-used targets big
- Group targets that are used together
 - Grouped toolbar buttons, menu items, etc.
- Place oftenly-used menu items on top of menu
- Use screen corners and edges



Principles to improve efficiency (cont'd)

- Use keyboard shortcuts and menu accelerators

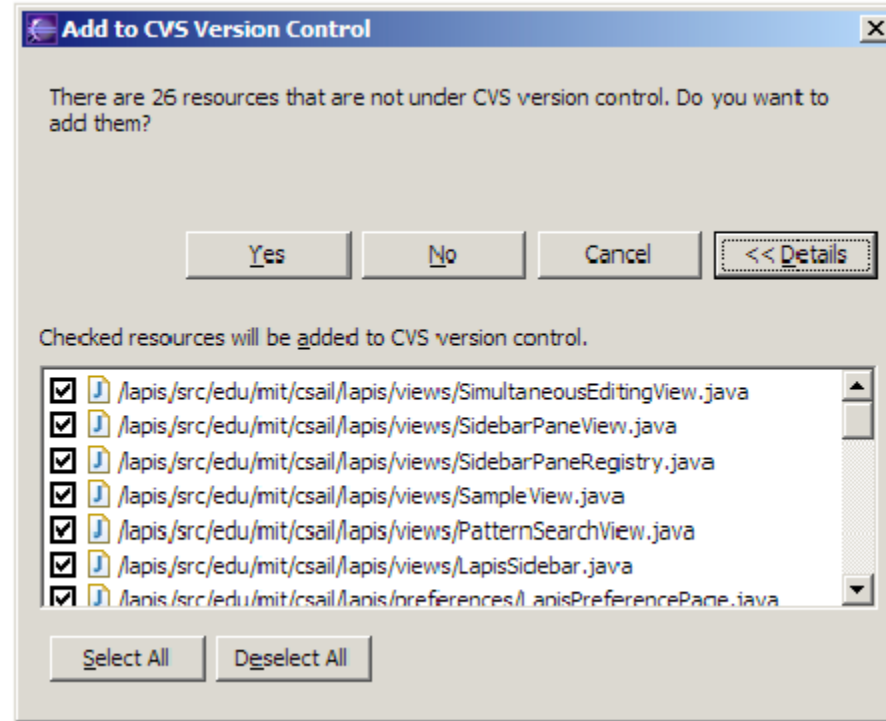


- Predefine a group of styles



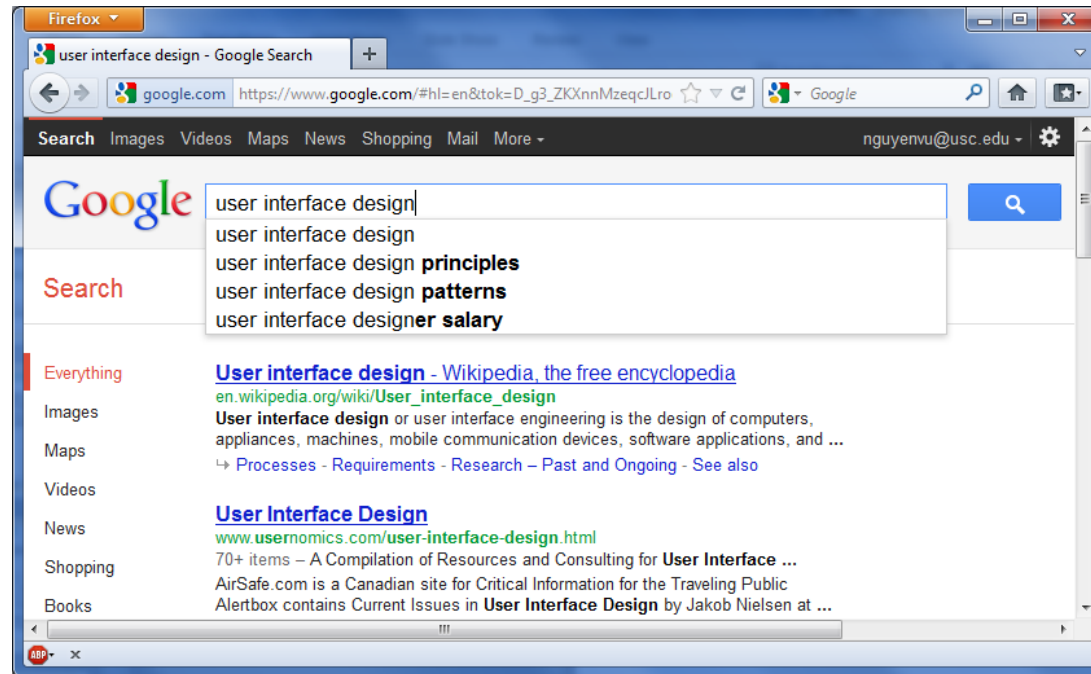
Principles to improve efficiency (cont'd)

- Aggregating and choose most common selections by default
- Use defaults



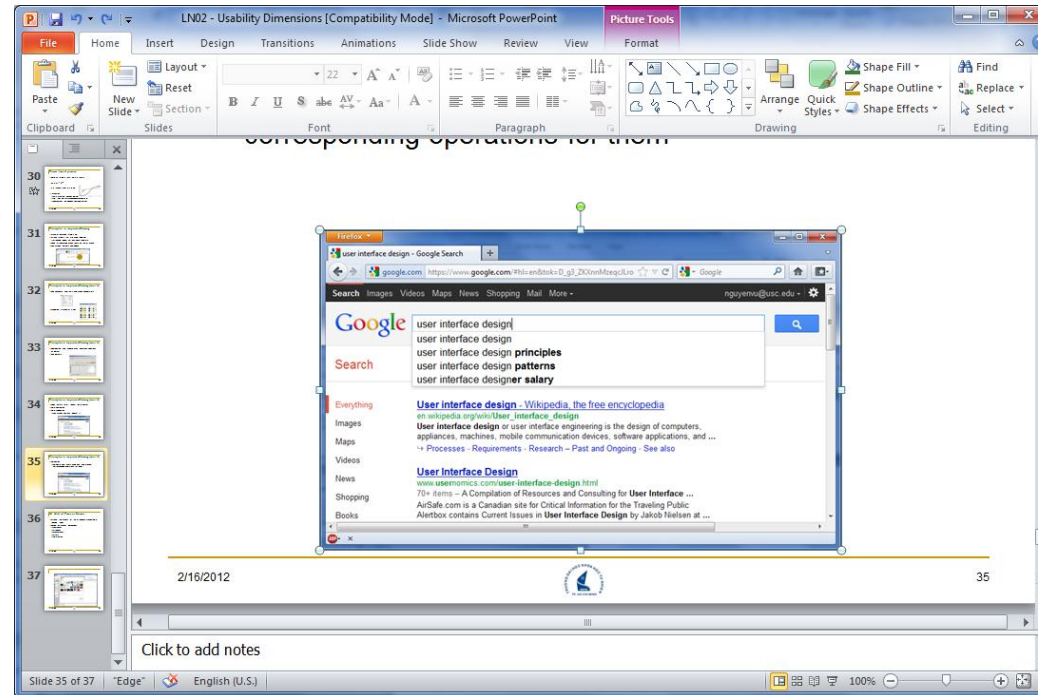
Principles to improve efficiency (cont'd)

- Keep history (e.g., recent files in Word)
- Auto completion
- Auto suggestion
 - This makes you lazy, doesn't it?



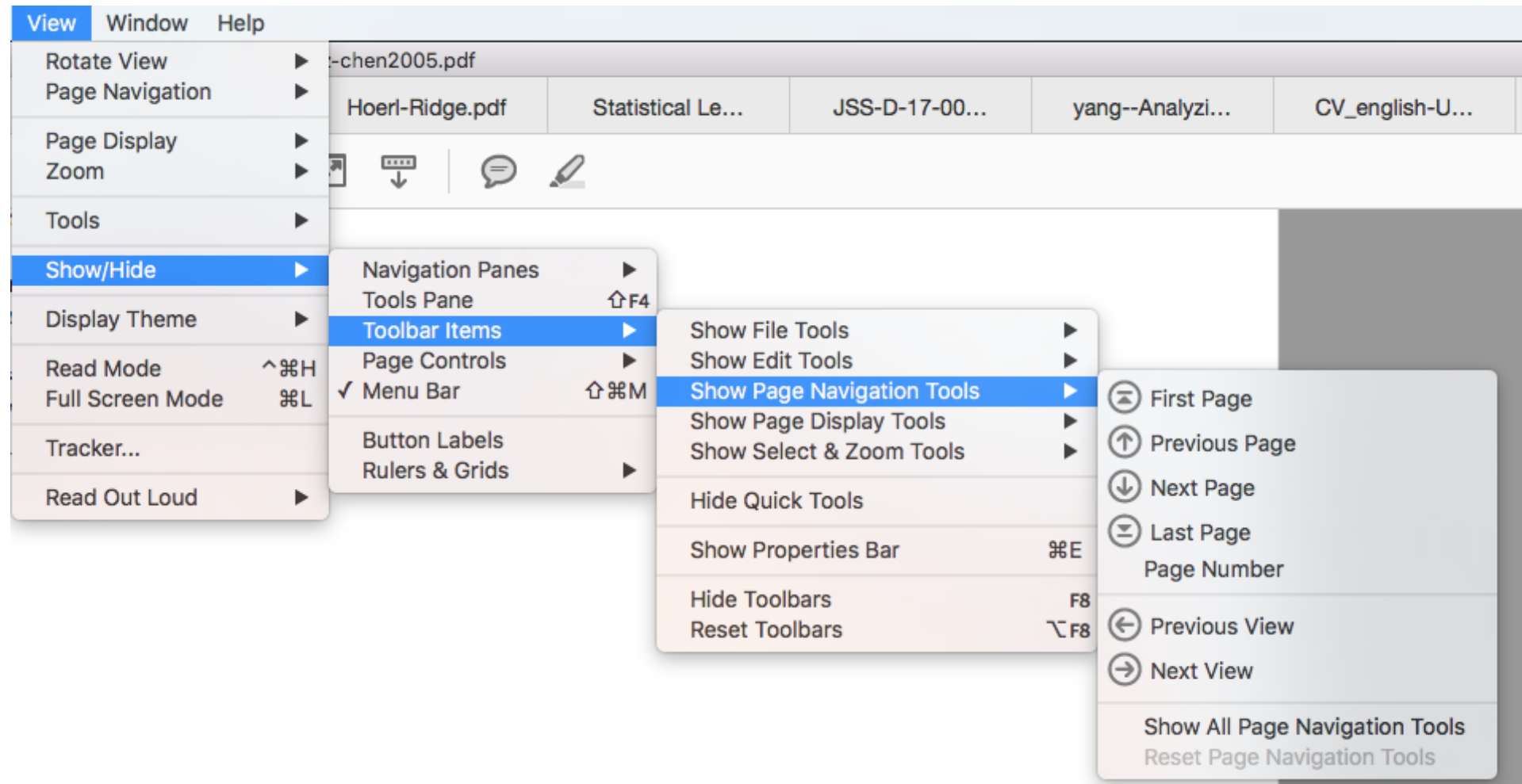
Principles to improve efficiency (cont'd)

- Anticipation
 - Anticipate what users will do next and present corresponding operations for them



Bad menu for efficiency

- Acrobat Reader for Mac



Mental models

- Craik (1943) described mental models as:
 - internal constructions of some aspect of the external world enabling predictions to be made
- Involves unconscious and conscious processes
 - images and analogies are activated
- Deep versus shallow models
 - e.g. how to drive a car and how it works

Mental models

- Users develop an understanding of a system through learning about and using it -- mental model:
 - How to use the system (what to do next)?
 - What to do with unfamiliar systems or unexpected situations (how the system works)
- People make inferences using mental models of how to carry out tasks

Designs should not
confuse/conflict users'
mental model



Interface/Interaction Metaphors

- A set of user interface *visuals, actions* and *procedures* that exploit *specific knowledge* that users *already have of other domains*.
- The purpose is to give the user *instantaneous knowledge* about *how to interact with the user interface*. They are designed to be *similar to physical entities* but also *have their own properties*



MDM



Logs

Metaphors

- Metaphor is a presentation of real-world in user interface
- Advantages
 - Highly learnable
 - Connect with user's existing model easily
- Problems
 - Hard to design metaphors that are appropriate
 - Potentially deceptive and misleading
 - May not be used consistently everywhere
 - Culturally dependent (localization issue)



Interface/Interaction Metaphors



Interface/Interaction Metaphors







Interface/Interaction Metaphors



Digital Finnish Driving License by HiQ

Interface/Interaction Metaphors



Shneiderman's Eight Golden Rules

- Strive for consistency
- Cater to universal usability
- Offer informative feedback
- Design dialogs to yield closure
- Prevent errors, rapid recovery
- Permit easy reversal of actions
- Support user control
- Reduce memory load

Conceptual model

- A conceptual model is the mental model that people carry of how something should be done
- We describe them in terms of
 - Core activities
 - Objects
 - Interface metaphors

FORMULATING A CONCEPTUAL MODEL

- What will the users be doing when carrying out their tasks?
- How will the system support these?
- What kind of interface metaphor?
- What kinds of interaction modes and styles to use?